"ALEXANDRU-IOAN CUZA" UNIVERSITY OF IAȘI

FACULTY OF COMPUTER SCIENCE



MASTER'S DISSERTATION

Attribute-Based Proxy Re-Encryption

proposed by

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Session: July, 2026

Scientific Supervisor

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Chapter 1

Introduction

1.1 Context and Motivation

The last two decades have brought a fundamental transformation in how data is stored, processed, and shared. The traditional model, in which information remains under the direct physical control of the owner, has been gradually replaced by outsourcing to cloud infrastructures, operated by third-party providers whose economic and operational interests do not always align with those of the data owner. This transition brings significant operational benefits—scalability, availability, reduction of infrastructure costs—but simultaneously introduces a structural security problem: once data leaves the physical control of the owner, guaranteeing confidentiality and granular access control can no longer rely on administrative mechanisms or trust in the service provider.

The relevant threat model in these environments is generally that of a semi-trusted (honest-but-curious) server: the cloud provider correctly executes the agreed protocols but cannot be assumed to refrain from examining the content of the stored data if it is possible to do so undetected. In this context, access control can no longer be delegated to an access policy applied at the server level, but must be guaranteed at the mathematical level, before the data leaves the owner, so that no unauthorized entity, including the cloud provider itself, can recover the plaintext information.

Attribute-Based Encryption (ABE) emerged as a direct response to this requirement. Unlike classical public-key cryptography, where a message is encrypted for a single recipient identity, ABE allows the expression of logical access policies over a set of attributes, where decryption succeeds if and only if the user possesses a com-

bination of attributes that satisfies the policy associated with the data. This capability to express complex access conditions—arbitrary conjunctions and disjunctions of attributes—directly within the structure of the ciphertext, without knowing in advance the concrete identities of the authorized recipients, makes ABE the natural tool for granular data sharing in cloud environments with large and dynamic user populations.

However, access control through ABE solves only a part of the problem. Real cloud environments also require a mechanism through which already granted decryption rights can be dynamically delegated, redirected, or restricted, without requiring the explicit decryption and re-encryption of the data by the owner. This capability is provided by proxy re-encryption (Proxy Re-Encryption, PRE), an independent cryptographic primitive, initially developed outside the ABE context. The convergence of the two directions, the expressiveness of access policies in ABE and the flexibility of delegation in PRE, produces Attribute-Based Proxy Re-Encryption (AB-PRE), the cryptographic primitive around which this work is structured. Depending on where the access policy resides, AB-PRE is instantiated in two dual architectural variants: Key-Policy AB-PRE (KP-ABPRE), in which the policy is embedded in the private key, and Ciphertext-Policy AB-PRE (CP-ABPRE), in which the policy is embedded in the ciphertext. The structural relationship between these two variants constitutes one of the central questions of the work.

1.2 Dissertation Objectives - Previous work

The central objective of this dissertation is twofold. On the one hand, the work aims to build a coherent and rigorous path through the mathematical and architectural foundations necessary for understanding AB-PRE, starting from basic cryptographic primitives and advancing systematically toward contemporary constructions. On the other hand, the work aims to answer a precise structural question left open in the literature: can the relationship between KP-ABE and CP-ABE, usually described only intuitively as an ”architectural inversion” between the place where the access policy resides and the place where the attributes reside, be made explicit at the level of the algebraic expressions that define the algorithms of the two schemes?

The answer to this question cannot be constructed in isolation, but requires a prior understanding of the entire evolutionary trajectory that the specialized literature

has traversed. Therefore, the work follows this trajectory in its historical and conceptual order: from identity-based cryptography (IBE), through its error-tolerant extension (Fuzzy IBE), to the two dual paradigms, KP-ABE and CP-ABE, and, finally, to their fusion with proxy re-encryption in the AB-PRE construction. Each step of this trajectory is motivated by a concrete limitation identified in the previous step, and traversing it is not merely an exercise in bibliographic synthesis, but the necessary foundation to be able to formulate precisely, in Chapter 5, the question regarding the algebraic relationship between KP-ABE and CP-ABE, and to correctly evaluate what exactly from the CP-ABPRE architecture inherits this relationship, and to what extent it propagates to the level of AB-PRE constructions with proxy re-encryption.

1.3 Contributions

The original contribution of this work, presented in Chapter 5, consists of a structural transformation $\mathcal{S}$ that algebraically derives the CP-ABE construction from the KP-ABE construction. Unlike the descriptive characterization present in the literature of the two paradigms as being "dual" to each other, this transformation is defined and verified explicitly at the level of the algebraic expressions that define the $Setup$, $Encrypt$, $KeyGen$, and $Decrypt$ algorithms of the two schemes, without resorting to a higher-level abstract formal apparatus. The central result establishes that the CP-ABE scheme can be obtained from the KP-ABE scheme through a composition of three elementary structural operations, applied to the algorithmic components of the latter, and precisely identifies which of the components of the CP-ABE scheme result necessarily from this transformation and which represent an independent addition, without a correspondent in the KP-ABE construction.

An equally important result, distinct from the transformation itself, concerns its limits: the algebraic identity thus established transports the correctness of the construction, but not its security guarantees, a distinction supported both by the formal structure of the transformation and by the fact that the security proofs published for the two schemes use mutually incompatible models and assumptions. Finally, the work explores to what extent the algebraic substrate thus identified is relevant for obtaining the KP-ABPRE and CP-ABPRE constructions, examining the relationship between the two architectural variants of AB-PRE and whether there exists a structural path analogous to $\mathcal{S}$ that directly links KP-ABPRE to CP-ABPRE (Figure 1.1).

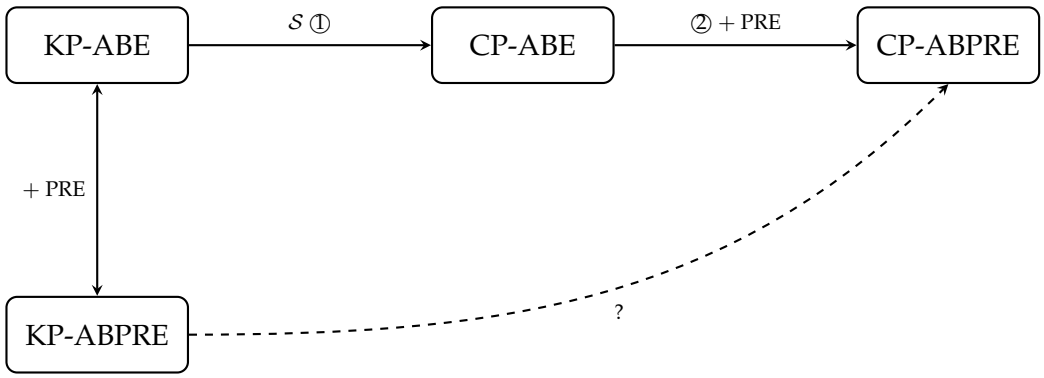


Figure 1.1: The structural relationship between ABE and AB-PRE constructions. Arrow ① represents the transformation $\mathcal{S}$ (the contribution of Chapter 5); arrow ② represents the fusion of CP-ABE with PRE (Chapter 4). The dashed arrow indicates the open question regarding the existence of a direct structural transformation from KP-ABPRE to CP-ABPRE.

1.4 Dissertation Structure

The remainder of this work is organized as follows.

Chapter 2 establishes the mathematical and cryptographic foundations necessary throughout the work: bilinear maps and their properties, the hardness assumptions used in the security proofs (BDH, DBDH and q-Parallel BDHE), access structures and linear secret sharing schemes (LSSS), as well as the three fundamental dimensions of the security space relevant for ABE and PRE: attack resistance, adversary power, and the proof model.

Chapter 3 traverses the evolutionary trajectory from identity-based cryptography to CP-ABE, passing through the limitations of IBE, through Fuzzy IBE as a first step towards treating identity as a set of attributes, and through the two dual architectures, KP-ABE and CP-ABE, concluding with a structural comparison between them that explicitly formulates the question resumed in Chapter 5.

Chapter 4 introduces proxy re-encryption as an independent primitive, analyzes its structural dimensions (bidirectionality vs. unidirectionality, single-hop vs. multi-hop, the proxy's trust model), and builds, through its fusion with CP-ABE, the CP-ABPRE architecture: its complete syntax, the specific security challenges (collusion resistance, master key security, the difficulty of achieving CCA security in the presence of a proxy) and a comparative analysis of the existing constructions in the literature.

Chapter 5 presents the original contribution of the work: the structural transfor-

mation $\mathcal{S}$ between KP-ABE and CP-ABE, its formal proof, the distinction between the transfer of correctness and the transfer of security, the discussion regarding the relevance of the identified algebraic substrate for the $ReKeyGen$ algorithm in CP-ABPRE, and an analysis of the structural relationship between the KP-ABPRE and CP-ABPRE constructions as instantiations of the same AB-PRE primitive.

Chapter 6 synthesizes the obtained results, discusses their limitations and outlines open research directions, including the question regarding the existence of a direct structural transformation from KP-ABPRE to CP-ABPRE.

Chapter 2

Cryptographic Foundations

This chapter establishes the theoretical and mathematical foundations necessary for understanding modern Attribute-Based Encryption (ABE) and Proxy Re-Encryption (PRE) schemes. In the context of cloud environments, where data is outsourced to semi-trusted entities, security can no longer rely exclusively on physical control or traditional reference monitors. Instead, protection must be guaranteed at the mathematical level [10]. To achieve advanced mechanisms, such as the flexible delegation of access rights and collusion resistance, modern cryptography relies on complex algebraic structures and mathematical problems whose resolution is considered computationally infeasible.

2.1 Mathematical Preliminaries

To implement fine-grained access control directly within the structure of a ciphertext, classical public-key schemes (such as RSA or standard elliptic curve cryptography) prove to be insufficient. These traditional mechanisms were designed to facilitate a one-to-one communication paradigm (a single private key decrypts a message intended for a single public key). In contrast, cloud environments necessitate a one-to-many dynamic, where the decryption process must evaluate a logical policy without compromising data confidentiality. This conceptual barrier was overcome through the adoption of mathematical primitives capable of evaluating hidden relationships within exponents, with bilinear maps serving as the foundation.

2.1.1 Bilinear Maps and Group Properties

The intuition behind bilinear maps (also known as pairings) is to provide an additional mathematical operation that allows combining two encrypted elements to reveal a relationship between them, without exposing their underlying raw values. In ABE and AB-PRE architectures, this property is exploited to "cancel out" the blinding factors (randomness) introduced during encryption by pairing them with complementary elements from the user's secret key, contingent upon the satisfaction of the access policy.

From a technical perspective, a bilinear map is defined over two multiplicative cyclic groups, G and G_T, both having the same prime order p. Let g be a generator of the base group G. A map $e : G \times G \to G_T$ is considered a cryptographic bilinear map if it simultaneously satisfies the following three fundamental conditions [11]:

1. **Bilinearity:** For any elements $u, v \in G$ and any scalars $a, b \in \mathbb{Z}_p$, the relation $e(u^a, v^b) = e(u, v)^{ab}$ is satisfied. This property represents the functional core of the bilinear mapping, as it allows for the manipulation and homogenization of the key and ciphertext components during the decryption process.

2. **Non-degeneracy:** The map must not send all pairs of elements to the identity element of the target group. In other words, $e(g, g) \neq 1_{G_T}$, which guarantees that the group G_T is properly generated and can be utilized effectively from a cryptographic standpoint.

3. **Computability:** There must exist an efficient, polynomial-time algorithm capable of computing the value $e(u, v)$ for any pair of elements in group G.

Although asymmetric maps also exist (where the input groups G_1 and G_2 are distinct), a large portion of foundational attribute-based encryption and re-encryption schemes utilize the symmetric variant due to the simplicity of their security proofs and implementation [12].

2.1.2 Hardness Assumptions: BDH, DBDH, and q-Parallel BDHE

Any modern cryptographic scheme anchors its security in a mathematical problem that is assumed to be computationally intractable. In the context of pairing-based cryptography, these problems are known as hardness assumptions and represent the

foundation upon which formal security proofs are built. If an adversary were to successfully break the scheme, the simulator used in the proof could leverage that capability to solve the underlying mathematical problem, which would contradict the hardness assumption and validate, by reduction, the security of the scheme [13].

The simplest starting point is the Bilinear Diffie-Hellman (BDH) problem. Let G be a multiplicative cyclic group of prime order p, with generator g, and let $e : G \times G \rightarrow G_T$ be a bilinear map. The BDH problem assumes that, given a tuple of public elements (g, g^a, g^b, g^c) for random scalars $a, b, c \in \mathbb{Z}_p$, it is computationally infeasible to compute the value $e(g, g)^{abc} \in G_T$ [14]. Intuitively, although an adversary knows the individual components g^a, g^b, and g^c, the hardness property of the discrete logarithm prevents the extraction of the exponents and, implicitly, the computation of their masked product in G_T.

A decisional variant, more frequently used in the security proofs of ABE schemes, is the Decisional BDH (DBDH) assumption. This does not require the actual computation of the value $e(g, g)^{abc}$, but rather the probabilistic distinction between it and a random element $Z \in G_T$. Formally, the advantage of any probabilistic polynomial-time adversary in distinguishing the tuple $(g, g^a, g^b, g^c, e(g, g)^{abc})$ from the tuple (g, g^a, g^b, g^c, Z) is negligible with respect to the security parameter λ [3, 10]. This assumption is sufficient for proving the security of KP-ABE schemes, because in these constructions the access structure is embedded within the secret key, while the ciphertext contains only the set of attributes, a relatively simple structure that does not impose excessive requirements on the simulator during the reduction [2].

However, in Ciphertext-Policy ABE (CP-ABE) schemes, the situation becomes significantly more complex. The access policy is now an integral part of the ciphertext, and during the security proof, the simulator must be able to embed an arbitrary policy, represented as an LSSS matrix M^*, directly into the public parameters of the scheme [13]. The problem that arises is that an attribute can be associated with multiple rows of the matrix M^*, generating ambiguity and algebraic conflicts if the simulator does not possess sufficient structural degrees of freedom. The DBDH assumption does not provide a sufficient number of interrelated elements to handle this complexity, which causes reductions based exclusively on DBDH for CP-ABE to impose severe restrictions on the size of the supported formulas or lead to an exponential increase in the size of the public parameters [3].

This limitation motivated the introduction of the q-Parallel Bilinear Diffie-Hellman

Exponent (q-Parallel BDHE) assumption, a parameterized and stronger assumption specifically tailored to the needs of CP-ABE with expressive LSSS policies [13]. Formally, let $\boldsymbol{y}$ be a vector of random scalars $b_1, \ldots, b_q \in \mathbb{Z}_p$, and let $a, s \in \mathbb{Z}_p$. The assumption states that, given an extended tuple of public elements consisting of:

$$g,\ g^s,\ \left(g^{a^i}\right)_{\substack{1\le i\le 2q \\ i\neq q+1}},\ \left(g^{s\cdot b_j}\right)_{1\le j\le q},\ \left(g^{a^i/b_j}\right)_{\substack{1\le i\le 2q \\ i\neq q+1 \\ 1\le j\le q}},\ \left(g^{a^i\cdot s\cdot b_k/b_j}\right)_{\substack{1\le i\le q \\ 1\le j,k\le q \\ k\neq j}}$$

it is computationally infeasible to distinguish the element $T = e(g,g)^{a^{q+1}\cdot s} \in G_T$ from a uniformly distributed random element in G_T [13, 10].

The intuition behind this rich structure is crucial for understanding CP-ABE security proofs. Unlike DBDH, which provides the adversary with a small, fixed set of elements, the q-Parallel BDHE assumption provides the simulator with an extended family of interrelated terms. This allows the simulator to assign different exponential components to each row of the matrix M^* independently, ensuring that the terms cancel out correctly during the simulation for authorized users, but not for the adversary [13]. In other words, the structural degree of freedom provided by the q-Parallel BDHE assumption is proportional to the complexity of the access policy that the scheme must support.

From a comparative perspective, there is a fundamental trade-off between the strength of the assumption used and the expressiveness of the resulting scheme. Simpler assumptions, such as DBDH, are theoretically preferable because they are better studied and less susceptible to future attacks. However, they constrain either the expressiveness of the policy (for example, limiting the scheme to AND gates with static attributes) or the efficiency of the parameters. The q-Parallel BDHE assumption, although stronger, remains non-interactive and has demonstrated sufficient empirical resilience to be adopted in CP-ABE and CP-ABPRE constructions with arbitrary LSSS policies, featuring constant-size public parameters and ciphertexts that scale linearly with the complexity of the formula [13, 15].

2.2 Access Structures and Secret Sharing

The cryptographic mechanisms presented in the previous section - bilinear maps and hardness assumptions - provide the raw mathematical tools required to construct ABE schemes. However, in order to utilize these tools for the purpose of fine-grained

access control, a formal bridge is necessary between the domain of access logic (who is permitted to decrypt and under what conditions) and the algebraic domain of exponents and cyclic groups. This bridge is provided by access structures and secret sharing schemes, which constitute the conceptual core of any ABE or AB-PRE construction.

2.2.1 Monotonic Access Structures and Access Trees

Before a message can be encrypted against a policy such as *"the user must possess the Doctor attribute and the Hospital-X attribute, or the Administrator attribute"*, the system must possess a precise mathematical formalism to describe which combinations of attributes are considered authorized. This description is captured by the notion of an access structure.

Let $\mathcal{U} = \{P_1, P_2, \ldots, P_n\}$ be the attribute universe of the system. An access structure $\mathbb{A}$ is a collection of non-empty subsets of $\mathcal{U}$, meaning $\mathbb{A} \subseteq 2^{\mathcal{U}} \setminus \{\emptyset\}$. The subsets included in $\mathbb{A}$ are called *authorized sets*, and the excluded ones are called *unauthorized sets* [2, 3]. Decryption is possible if and only if the user's set of attributes forms an authorized set with respect to the access structure associated with the ciphertext.

An essential property, present in all CP-ABE and CP-ABPRE constructions studied in this work, is *monotonicity*. An access structure $\mathbb{A}$ is monotone if for any two subsets $B, C \subseteq \mathcal{U}$, the condition $B \in \mathbb{A}$ and $B \subseteq C$ implies $C \in \mathbb{A}$ [16, 2]. Intuitively, monotonicity guarantees that possessing additional attributes cannot revoke an already existing access right. Although non-monotone structures, which also allow for negative conditions of the type *"the user must not possess attribute X"*, have been studied in the literature [17], they introduce significant complexity in both the construction and the security proofs. Therefore, foundational CP-ABE and CP-ABPRE schemes typically operate with monotone access structures, an assumption maintained in this work as well.

An intuitive and frequently used representation of monotone access structures is that of *access trees* [2, 1]. An access tree $\mathcal{T}$ is a directed tree in which the interior nodes represent threshold gates, and the leaf nodes are associated with the attributes of the system. Each interior node x is characterized by two parameters: the number of its children num_x and a threshold value k_x, where $1 \leq k_x \leq num_x$. The most common special cases are the *OR* logic gate, corresponding to a threshold $k_x = 1$, and the *AND* logic gate, corresponding to a threshold $k_x = num_x$. Each leaf node x is associated with

a specific attribute and has, by convention, $k_x = 1$.

The evaluation of an access tree against a set of attributes S is performed recursively, from the leaves to the root. A leaf node is satisfied if its associated attribute is found in S. An interior node x is satisfied if at least k_x of its children are satisfied. The tree as a whole is satisfied if its root is satisfied [2]. This graphical representation facilitates the intuitive understanding of complex policies and served as the basis for the first CP-ABE constructions proposed by Bethencourt et al. [3]. However, from an algebraic perspective, access trees present a limitation: their direct translation into ciphertext components leads to key and ciphertext structures whose size grows proportionally with the depth and width of the tree, negatively impacting the efficiency of the scheme.

This practical limitation motivated the transition to an alternative representation, more suitable for formal security proofs and efficient constructions: linear secret sharing schemes.

2.2.2 Linear Secret-Sharing Schemes (LSSS)

Linear Secret-Sharing Schemes (LSSS) represent the unifying algebraic framework that enables the translation of any monotone access structure into a mathematical format directly operable within the exponents of cyclic groups [16, 10]. Unlike access trees, which are constructed recursively and evaluated structurally, an LSSS expresses the same logical content through a matrix and a labeling function, thereby providing precise algebraic control over the secret sharing and reconstruction process.

Formally, a secret-sharing scheme Π over a set of attributes $\mathcal{U}$ is linear over $\mathbb{Z}_p$ if it is defined by a pair (M, ρ), where M is a matrix with ℓ rows and n columns over $\mathbb{Z}_p$, and $\rho : [\ell] \to \mathcal{U}$ is a labeling function that associates each row i of the matrix with an attribute from $\mathcal{U}$ [13, 10]. It is important to note that the function ρ is not necessarily injective: the same attribute can label multiple distinct rows of the matrix M, a fact with direct implications in the security proofs for CP-ABE, as discussed in the previous section in the context of the q-Parallel BDHE assumption.

The mechanism for sharing a secret $s \in \mathbb{Z}_p$ operates as follows. A random column vector $v = (s, r_2, r_3, \ldots, r_n)$ is constructed, where $r_2, \ldots, r_n \in \mathbb{Z}_p$ are chosen uniformly and independently. The vector of shares is defined as the matrix product $M \cdot v \in \mathbb{Z}_p^\ell$. The share corresponding to row i is $\lambda_i = (M \cdot v)_i$ and belongs to the attribute $\rho(i)$

[13, 10]. By construction, each share λ_i is a linear combination of the secret s and the random values $r_2, \ldots, r_n$, which means that no individual share reveals information about s in the absence of the others.

The central property of LSSS schemes, which makes them usable in CP-ABE constructions, is linear reconstruction. If a user possesses an authorized set of attributes $S \in \mathbb{A}$, let $I = \{i : \rho(i) \in S\}$ be the subset of rows labeled with attributes from S. The reconstruction property guarantees that the target vector $(1, 0, \ldots, 0)$ is in the linear span of the rows of the submatrix M_i [16, 10]. Therefore, there exist reconstruction constants $\{w_i \in \mathbb{Z}_p\}_{i \in I}$, computable in polynomial time, such that: $\sum_{i \in I} w_i \lambda_i = s$. Conversely, for any unauthorized set $S \notin \mathbb{A}$, the vector $(1, 0, \ldots, 0)$ is not in the linear span of the corresponding rows, which implies the impossibility of reconstructing the secret s from the available shares [16, 13]. The relevance of this mechanism for CP-ABE and CP-ABPRE schemes is direct and profound. In the encryption algorithm, the secret s plays the role of a masking exponent: the message is blinded by multiplication with $e(g, g)^{\alpha s}$ for a parameter α from the system setup, and the shares λ_i are embedded within the ciphertext components associated with each attribute in the access policy. In the decryption algorithm, a user with authorized attributes applies pairing operations between the ciphertext components and their secret key components, obtaining the values $e(g, g)^{\lambda_i \cdot (\text{key term})}$ for each $i \in I$. By applying the reconstruction constants w_i in the exponent and combining the results, the masking factor cancels out completely, enabling the recovery of the message [13, 15].

A critical aspect, specific to cloud environments and directly related to collusion resistance, is how ABE schemes prevent unauthorized users from combining their shares. Unlike classical secret sharing schemes, in which any k out of n participants can reconstruct the secret through cooperation, in ABE the combination of keys belonging to different users must not allow decryption, even if together they would satisfy the policy [2, 3]. This is technically achieved by introducing a unique randomization factor per user in the key generation process: the shares within a user's key are blinded with a random element t chosen independently for each key issuance, such that the key components of different users cannot be algebraically combined in a coherent manner [2, 15]. This property, although simple in its formulation, has fundamental implications in the security proofs of CP-ABPRE schemes, as will be discussed in the following chapters.

2.3 Security Models

Security models define what it means for a construction to be secure. Without a precise formal model, a cryptographic scheme cannot be rigorously evaluated, and security claims remain at an intuitive level. In the context of CP-ABE and CP-ABPRE, the choice of the security model has direct implications on the expressiveness of the scheme, the efficiency of the construction, and the strength of the guarantees provided. This section establishes the three fundamental dimensions of the security space relevant to this work: the level of resistance against attacks (CPA vs. CCA), the adversary's power in choosing the target (selective vs. adaptive), and the nature of the assumptions used in the proofs (random oracle model vs. standard model).

2.3.1 CPA vs. CCA Security

The most widely used formal framework for evaluating the security of public-key encryption schemes is the indistinguishability game, played between a probabilistic polynomial-time adversary and a simulator. The adversary receives a challenge ciphertext and must determine which of two messages of their choosing was encrypted, without being able to do so with a probability significantly greater than that of random guessing.

The fundamental level of security within this framework is IND-CPA (Indistinguishability under Chosen-Plaintext Attack), also known as semantic security. In an IND-CPA game, the adversary can encrypt any messages of their choice using the public keys of the system, but does not have access to any decryption oracle. Formally, the scheme is IND-CPA secure if for any efficient adversary $\mathcal{A}$, its advantage defined as: $\mathsf{Adv}_{\mathcal{A}}^{\mathrm{CPA}} = \left|\Pr[\mathcal{A} \text{ wins}] - \frac{1}{2}\right|$ is negligible in the security parameter λ. The first CP-ABE schemes, including the seminal construction by Bethencourt et al. [3], exclusively targeted IND-CPA security, as this represents the minimum acceptable level for confidentiality.

In cloud environments, however, adversaries are not passive. An active attacker can intercept ciphertexts in transit, subtly modify them, and observe the system's behavior following decryption attempts, thereby obtaining partial information about the original message. This scenario is captured by the IND-CCA (Indistinguishability under Chosen-Ciphertext Attack) model, in which the adversary is additionally granted access to a decryption oracle $\mathcal{O}_{\mathrm{dec}}$, to which they can submit arbitrary ciphertexts and

observe the results [18]. The classical distinction between CCA1, where the oracle is accessible only before receiving the challenge ciphertext, and CCA2, where the oracle remains adaptively accessible throughout the game, is particularly relevant in the context of PRE, where the rights delegation structure creates additional attack surfaces [15]. The only restriction imposed on the adversary in the CCA2 game is that they cannot query the decryption oracle with the challenge ciphertext C^* itself, a necessary condition to prevent a trivial win.

The extension of the CCA model to CP-ABPRE schemes introduces additional complexity, as the presence of the proxy generates new oracles and new attack vectors. A complete security game for CP-ABPRE provides the adversary with access to a re-encryption key generation oracle $\mathcal{O}_{rk}$, a re-encryption oracle $\mathcal{O}_{re}$, a decryption oracle for level 2 (original) ciphertexts $\mathcal{O}_{d2}$, and a decryption oracle for level 1 (re-encrypted) ciphertexts $\mathcal{O}_{d1}$ [15, 19]. To prevent a trivial win, the model forbids querying the decryption oracles not only with the challenge ciphertext C^*, but also with any derivative thereof, meaning any ciphertext obtained through the legitimate re-encryption of C^* under a policy for which the adversary possesses or can obtain the corresponding key [15].

A fundamental difficulty, specific to proxy re-encryption, is the incompatibility between CCA security and the transformation functionality of the proxy. In classical cryptography, the CHK (Canetti, Halevi, Katz) transformation [20] allows the conversion of any IND-CPA scheme into an IND-CCA one by attaching a one-time signature to the ciphertext, guaranteeing its integrity. Naively applied to a PRE system, this transformation becomes self-destructive: the proxy must structurally modify the ciphertext in the re-encryption process, and any legitimate modification invalidates the attached signature, blocking subsequent decryption [7]. This fundamental tension motivated the introduction of an intermediate security model, RCCA (Replayable Chosen-Ciphertext Attack) [21], in which the adversary is permitted to obtain decryptions of ciphertexts that represent harmless variants of the challenge, meaning variants that do not alter the underlying message. RCCA naturally captures the semantics of re-encryption, because precisely this invariance of the message with respect to the proxy transformation is the central property of PRE schemes [7, 15].

2.3.2 Selective vs. Adaptive Security

A second dimension of the security space concerns the moment when the adversary chooses their attack target. This distinction, seemingly procedural, has profound consequences on the usable proof techniques and the strength of the obtained guarantees.

In the selective security model, the adversary is required to declare the target access structure $\mathbb{A}^*$, or in the case of KP-ABE the target attribute set, in the initial phase of the game, before receiving the public parameters of the system and before making any queries [1, 2]. This early commitment provides the simulator with a crucial advantage: knowing $\mathbb{A}^*$ from the start, it can utilize a partitioning strategy, injecting the components of the hard mathematical problem (e.g., the elements of the DBDH tuple) directly into the public parameters associated with the attributes in $\mathbb{A}^*$, simultaneously handling the queries for attributes outside $\mathbb{A}^*$ with values it explicitly controls [1, 13]. The result is a clean and relatively simple reduction, but one that does not reflect the behavior of a realistic adversary: in cloud environments, an attacker first observes the system parameters, collects information about the available keys, and only subsequently dynamically chooses the most vulnerable target [13].

The adaptive security model (full security or adaptive security) removes this artificial restriction, allowing the adversary to choose $\mathbb{A}^*$ adaptively during the challenge phase, after having seen the public parameters and having performed an arbitrary number of key queries [10, 13]. This correctly formalizes a real-world adversary and represents the standard objective of any modern cryptographic scheme. The technical difficulty is that the partitioning strategy is no longer applicable: the simulator does not know in advance which attributes will constitute the target, therefore it cannot properly inject the hard problem without compromising the simulation of key queries.

The elegant solution to this problem was provided by Lewko and Waters [10] through the Dual System Encryption technique. The central idea consists of expanding the space of keys and ciphertexts with an additional form, termed semi-functional. Normal keys and ciphertexts behave exactly as in the original scheme: normal keys decrypt normal ciphertexts, and normal keys also decrypt semi-functional ciphertexts. In contrast, a semi-functional key applied to a semi-functional ciphertext produces an additional residual term that causes decryption to fail [10]. The security proof progresses through a sequence of hybrid games in which the challenge ciphertext and the issued

keys are gradually replaced with their semi-functional variants, each transition being computationally indistinguishable from the previous one under the chosen hardness assumption. At the end of the sequence, the adversary faces a game in which all keys are semi-functional and, therefore, are of no use to them in distinguishing the challenge ciphertext [10]. This technique enabled the achievement of the first adaptively secure CP-ABE schemes in the standard model, eliminating the need for the artificial commitments of the selective model.

2.3.3 Random Oracle Model vs. Standard Model

The third dimension of the security space concerns the nature of the assumptions used in the proofs and, in particular, how cryptographic hash functions are treated. This choice has direct implications on the strength of the theoretical guarantees and the complexity of the resulting constructions.

In the Random Oracle Model (ROM), cryptographic hash functions are idealized as completely random oracles: for any input, the oracle returns an output chosen uniformly and independently from the output space, with the sole constraint that identical inputs produce identical outputs [22]. This idealization dramatically simplifies the constructions and the proofs, allowing the simulator to program the responses of the hash oracle in a convenient manner to embed components of the hard mathematical problem. Many early CP-ABE and CP-ABPRE schemes, including constructions by Luo et al. [9] and some variants by Liang et al. [7], operated in the ROM, thus achieving increased efficiency and more accessible proofs.

The fundamental limitation of the ROM is that real-world hash functions, such as SHA-256 or SHA-3, are deterministic functions with a precise internal structure, not ideal random oracles. Canetti, Goldreich, and Halevi [23] demonstrated that there exist schemes that are provably secure in the ROM, but completely insecure under any concrete implementation of the hash function. Therefore, a proof in the ROM constitutes a heuristic guarantee, not an absolute formal guarantee.

The standard model (Standard Model, SM) eliminates this idealization: the proofs rely exclusively on established mathematical assumptions, such as DBDH or q-Parallel BDHE, without any additional assumption regarding the behavior of hash functions or other primitives [13, 10]. Achieving CCA and/or adaptive security in the standard model for CP-ABPRE schemes is technically significantly more difficult and requires

advanced techniques: strongly unforgeable one-time signatures to ensure the integrity of the ciphertext, composite order groups with orthogonal subgroups to separate normal components from semi-functional ones, or combinations of these techniques with dual system arguments [10, 15]. The cost in terms of computational complexity and parameter size is significantly higher compared to ROM-based schemes, but the guarantees provided are the only acceptable ones for critical applications in cloud environments with active adversaries.

Table 2.1 comparatively summarizes the three dimensions of the security space discussed in this section, indicating the implications and relevant trade-offs for the constructions analyzed in the following chapters.

Table 2.1: Dimensions of the security space in CP-ABE and CP-ABPRE

Dimension	**Basic level**	**Advanced level**	**Technique / Transition cost**
Attack resistance	IND-CPA	IND-CCA2	RCCA as an achievable intermediate level; full CCA2 requires OTS or NIZK techniques
Adversary power	Selective	Adaptive (full)	Dual System Encryption [10]; hybrid arguments in the standard model
Proof model	Random Oracle Model	Standard Model	Composite order groups, extended public parameters, one-time signatures

These three dimensions are not independent: the strongest guarantees—adaptive CCA2 security in the standard model—are also the most costly to achieve and represent the active frontier of research in the field of AB-PRE.

Chapter 3

Attribute-Based Encryption: From IBE to CP-ABE

The previous chapter established the fundamental mathematical tools upon which modern cryptographic access control schemes are built: bilinear maps, hardness assumptions, and LSSS schemes. Before analyzing CP-ABE and CP-ABPRE constructions in detail, it is necessary to understand the historical and conceptual trajectory that led to their emergence. This evolution is not strictly linear, but rather represents a succession of responses to concrete limitations identified in previous schemes, with each step introducing an additional degree of expressiveness and flexibility in data access control.

The starting point of this evolution is identity-based cryptography, which represented the first significant step beyond traditional public key infrastructures. From there, the transition to attribute-based encryption was motivated by practical requirements in data sharing environments, and subsequent refinement led to the separation between the access policy and the user's key — the fundamental distinction between KP-ABE and CP-ABE, which will be analyzed in detail in Sections 3.3 and 3.4.

3.1 Identity-Based Encryption and Its Limitations

Traditional Public Key Infrastructures (PKI) solve the key distribution problem through digital certificates issued by Certificate Authorities (CA). Although functional, these infrastructures introduce significant operational complexity: before a sender can encrypt a message for a recipient, they must obtain and verify their digital certificate,

assuming the existence of a trusted CA, a functional revocation protocol, and synchronization among all participants. In dynamic and distributed environments, such as cloud systems or sensor networks, this reliance on infrastructure becomes a significant practical obstacle.

The concept of Identity-Based Encryption (IBE) was proposed by Shamir [24] precisely as a response to this complexity. The central idea is radical in its simplicity: a user's public identity, an email address, a name, an identification number, acts directly as a public key. A sender can thus encrypt a message intended for `bob@spital.ro` without previously holding any certificate or separate public key, using exclusively the character string of the identity as an input parameter in the encryption algorithm. The complexity of PKI is replaced by a central trusted entity, called the Private Key Generator (PKG), which holds a master secret key and derives from it the private keys corresponding to each identity upon request [24].

The practical realization of this concept remained open for almost two decades, until Boneh and Franklin [14] proposed the first efficient IBE construction, based precisely on the bilinear maps introduced in Section 2.1. Their scheme, known by the acronym BF-IBE, utilizes the bilinearity property to mathematically link the user's identity to their private key and the ciphertext, allowing decryption if and only if the private key was derived for the identity used during encryption. Boneh and Franklin proved the security of the scheme in the random oracle model under the BDH assumption, establishing a methodological precedent for all subsequent pairing-based constructions [14]. However, IBE presents two fundamental structural limitations that make it unsuitable for flexible data sharing scenarios in cloud environments.

The first limitation is identity rigidity. In the BF-IBE scheme, the identity is treated as an exact string of characters: encryption is performed against a precise identity, and decryption succeeds if and only if the private key corresponds to that exact same identity. Any discrepancy, whether a typographical error, a formatting change, or a semantically close variant, causes decryption to fail completely [1]. This rigidity is acceptable in point-to-point communication scenarios, but becomes problematic in any context where the user's identity is expressed through variable or approximate credentials, such as biometric data or organizational roles.

The second limitation, more profound from the perspective of cloud applications, is the coarse granularity of access control. IBE was designed to send data to a single targeted identity. In a real-world scenario of medical data sharing in the cloud, the

data owner wants to express policies such as *"any cardiologist from Hospital X"* or *"senior audit committee members"* — policies that target categories of users defined by attributes, not individual identities. Traditional IBE cannot express these policies directly: the only alternatives are the repeated encryption of the same message for each individual identity in the targeted category, or the issuance of the same private key to all members of the category, both variants destroying either the efficiency or the security properties of the system [2].

These two limitations together define the problem to which attribute-based cryptography will provide the solution: there is a need for a mechanism where the addressing unit is no longer the individual identity, but a set of attributes or a logical policy, allowing for fine-grained and expressive access control without sacrificing security or efficiency.

3.2 Fuzzy IBE: The Bridge to Attribute-Based Approaches

The first step toward resolving these limitations was taken by Sahai and Waters [1] by introducing the concept of Fuzzy Identity-Based Encryption (Fuzzy IBE). Their fundamental contribution consists of changing the perspective on identity: instead of being treated as a unique and rigid string of characters, identity is now viewed as a set of descriptive attributes, denoted as $\omega \subseteq \mathcal{U}$, where $\mathcal{U}$ is the universe of system attributes. This conceptual change allows the introduction of a similarity metric between identities: a private key generated for the identity ω can decrypt a ciphertext encrypted against the identity ω' if and only if the overlap between the two sets of attributes exceeds a minimum threshold d established at system initialization: $|\omega \cap \omega'| \geq d$. The parameter d is public and fixed for the entire system instance [1]. This property, termed error tolerance or fuzzy matching, directly solves the first limitation of IBE: small variations in identity no longer block decryption, as long as a sufficient number of attributes remain common.

The initial motivation for this construction was biometric cryptography. Biometric data — fingerprints, iris scans, facial recognition — inherently exhibit variability between successive measurements of the same individual, due to sensor noise, lighting conditions, or minor physiological variations [1]. A classical IBE system would fail to decrypt whenever the biometric sample at the time of decryption does not exactly match the one recorded at encryption. Fuzzy IBE solves this problem through

its tolerance for partial overlap, while maintaining collusion resistance for users with completely disjoint attribute sets.

The technical implementation of this mechanism is based on Shamir's secret sharing [25] applied in the exponent of bilinear groups. In the key generation algorithm, the PKG associates with each user with identity ω a random polynomial $q(x)$ of degree $d-1$ over $\mathbb{Z}_p$, where the secret $s = q(0)$ is the value that the scheme attempts to recover during decryption. The private key component corresponding to the attribute $i \in \omega$ is derived by evaluating the polynomial at point i, combined with the system parameters [1]. In the decryption algorithm, if the user possesses at least d attributes in common with the identity against which the encryption was performed, they can apply Lagrange interpolation in the exponent, exploiting the bilinearity of the map e, to recover $e(g,g)^{\alpha s}$ and decrypt the message.

Collusion resistance is structurally guaranteed by the independent randomness of the polynomial $q(x)$ assigned to each user. Two users with identities ω_1 and ω_2, each having fewer than d attributes in common with the encryption identity, cannot combine their key components to reach the required threshold, because their polynomials are mathematically incompatible: the components generated based on $q_1(x)$ cannot be interpolated together with those generated based on $q_2(x)$ [1]. This is the first formal realization of the collusion resistance property within ABE, a property that will remain central to all the constructions analyzed in this work.

However, Fuzzy IBE remains limited to a single access structure: the threshold gate with a fixed threshold d. All attributes are treated as equivalent, and the only evaluated condition is the number of common attributes, without any possibility of expressing complex logical relationships such as selective conjunctions, disjunctions, or negations [2]. A data owner in a cloud environment cannot express policies of the type *"Doctor AND (Cardiologist OR Surgeon) AND Hospital-X"*, which is exactly the type of granular policies required in real-world applications.

This limitation in expressiveness constituted the direct motivation for the next step in the evolution of ABE: the introduction of arbitrary access structures represented by access trees and, subsequently, by LSSS matrices. The answer to this challenge came through two major contributions, KP-ABE and CP-ABE, which differ fundamentally in the way they associate the access policy with the user's key and the ciphertext, respectively, as will be analyzed in the following sections.

3.3 Key-Policy ABE (KP-ABE)

Fuzzy IBE demonstrated that treating identity as a set of attributes opens new possibilities in cryptographic access control. However, its limitation to a single threshold gate with a fixed parameter remained a significant obstacle for real-world applications. The answer to this limitation came through the contribution of Goyal et al. [2], who fundamentally generalized the secret sharing mechanism of Fuzzy IBE, replacing the uniform threshold gate with arbitrary access structures represented by access trees. The resulting scheme, termed Key-Policy Attribute-Based Encryption (KP-ABE), represented the first truly expressive ABE system and established the base architecture that all subsequent ABE constructions would refine or invert.

3.3.1 System Architecture: Keys Embed Policies, Ciphertexts Embed Attributes

The organizing principle of KP-ABE is a clear separation of responsibilities between the sender and the key distribution authority, a separation that determines the entire architecture of the scheme. The sender who encrypts a message has a passive role from the perspective of the access policy: they associate with the ciphertext a set of descriptive attributes $\gamma \subseteq \mathcal{U}$, which describe the context or content of the encrypted data — for example, {`Network`, `Log`, `Source-IP-X`, `High-Priority`} for an audit system [2]. The sender does not specify and does not necessarily know who will be able to decrypt the message; this decision belongs entirely to the central authority.

The authority, in turn, embeds the access logic directly into the private key of each user, in the form of an access tree $\mathcal{T}$ in the sense defined in Section 2.2.1. The interior nodes of this tree are generalized threshold gates, and the leaves correspond to attributes in $\mathcal{U}$. In the key generation algorithm, the authority applies a top-down secret sharing procedure: starting from the root of the tree, the master secret necessary for decryption is distributed polynomially along the branches, such that each interior node with a threshold k_x out of num_x children generates a polynomial of degree $k_x - 1$ and transmits evaluations of it to each child [2]. The process continues recursively down to the leaf level, where each leaf associated with the attribute i receives a key component derived from the polynomial value accumulated at that node, combined with the system's public parameters corresponding to the attribute i.

The decryption algorithm works in the opposite direction, bottom-up. The user first checks whether the attribute set γ of the ciphertext satisfies the tree $\mathcal{T}$ of their key. For each satisfied leaf, meaning for each attribute in γ that appears in $\mathcal{T}$, the user applies a pairing operation between the corresponding component from the ciphertext and the corresponding component from their private key, obtaining a partial element in G_T. These partial elements are combined recursively through Lagrange interpolation in the exponent, moving up from the leaves to the root of the tree: at each interior node with a threshold k_x, a user who satisfies at least k_x children can reconstruct the contribution of that node by applying the appropriate Lagrange coefficients [2]. If the tree is entirely satisfied, the process culminates with the recovery of the masking factor $e(g, g)^{\alpha s}$ at the root, allowing the decryption of the message.

A distinctive structural property of KP-ABE, absent from Fuzzy IBE, is the capability of delegating access rights. Because the access policy is embedded in the private key and not in the ciphertext, a user with the key for the structure $\mathcal{T}$ can derive a new private key for a more restrictive structure $\mathcal{T}'$, provided that $\mathcal{T}'$ logically implies $\mathcal{T}$, meaning any attribute set that satisfies $\mathcal{T}'$ also satisfies $\mathcal{T}$ [2]. Technically, delegation is achieved by increasing the thresholds in the relevant interior nodes and re-randomizing the key components, such that the delegated key reveals no additional information compared to the original key. This property makes KP-ABE a natural tool for constructing hierarchical access control systems, in which higher-level authorities can delegate granular rights to subordinates without compromising global security.

3.3.2 Security Properties and Limitations

From the perspective of formal security, the KP-ABE scheme proposed by Goyal et al. [2] is proven to be IND-CPA secure in the selective security model (Selective-Set), described in Section 2.3.2. In the corresponding security game, the adversary commits to the target attribute set γ^* before receiving the public parameters, and the proof reduces to the BDH assumption through the partitioning strategy: the simulator injects the components of the BDH tuple into the public parameters associated with the attributes in γ^* and responds to key queries for structures not satisfied by γ^* using values that it explicitly controls [2]. The reduction is clean and efficient precisely because the target set is known from the start, which allows the simulator to anticipate exactly where it must place the components of the hard problem.

The collusion resistance property is guaranteed through the same mechanism identified in Fuzzy IBE, but now generalized to arbitrary access structures. Each execution of the key generation algorithm uses a completely new set of random polynomials, independent of all previous executions. The key components of two different users are thus linked to mathematically incompatible polynomials, and their combination cannot produce a coherent value that satisfies the access structure [2]. Formally, even if user A with structure $\mathcal{T}_A$ and user B with structure $\mathcal{T}_B$ could together cover the necessary attributes, they cannot combine their keys to decrypt a ciphertext that neither can decrypt individually.

Subsequent research focused on eliminating the selective restriction, considered artificial in relation to the behavior of real-world adversaries. Lewko et al. [10] extended the Dual System Encryption technique to KP-ABE, obtaining the first adaptively secure KP-ABE scheme in the standard model, without the idealization of the random oracle. This extension confirmed that the conceptual separation between keys with normal structures and those with semi-functional structures, introduced in Section 2.3.2, applies naturally in the context of KP-ABE as well, not just CP-ABE.

However, KP-ABE presents a fundamental limitation that makes it unsuitable for typical data sharing scenarios in the cloud. The limitation is not mathematical in nature, but architectural: in KP-ABE, the sender does not control the access policy. They tag the ciphertext with descriptive attributes, but the decision regarding who can decrypt belongs exclusively to the authority that issued the keys. The owner of a document stored in the cloud cannot specify at the moment of encryption that only *cardiologists from Hospital X* are allowed to access that document; this decision must be anticipated by the authority at the time of key issuance, independent of context [13, 2]. The model is suitable for systems with fixed and well-defined roles, such as the log audit systems described by Goyal et al., but it becomes rigid and counter-intuitive in any scenario where the data owner wishes to define the access policy dynamically, depending on the sensitivity of each individual file.

The second limitation, present in the basic constructions, concerns the monotonicity of the supported access structures. The original KP-ABE schemes did not support negative conditions of the `NOT Attribute` type, because the inclusion of negations broke the monotonicity property necessary in the security proofs. Ostrovsky et al. [17] addressed this problem by introducing a non-monotonic KP-ABE, utilizing revocation polynomials to express negative constraints without resorting to costly workarounds

such as dummy attributes, but at the price of additional complexity in both the construction and the security proof.

These two limitations, the lack of sender control over the policy and the monotonicity of the structures, constituted the direct motivation for inverting the KP-ABE architecture, resulting in the Ciphertext-Policy ABE paradigm, analyzed in the following section.

3.4 Ciphertext-Policy ABE (CP-ABE)

The fundamental limitation of KP-ABE identified in the previous section, namely that the sender does not control the access policy, is not a consequence of the underlying mathematics, but of an architectural choice: the policy was placed in the key, and the attributes in the ciphertext. Inverting this choice produces a completely different paradigm, more natural for cloud data sharing scenarios, in which the data owner and the sender are usually the same entity and wish to decide dynamically, at the moment of encryption, who can access each individual resource. This architectural inversion defines Ciphertext-Policy Attribute-Based Encryption (CP-ABE), introduced by Bethencourt, Sahai, and Waters [3] and formally refined by Waters [13].

3.4.1 System Architecture: Ciphertexts Embed Policies, Keys Embed Attributes

In CP-ABE, the roles of the two main components of the system are explicitly inverted compared to KP-ABE. The central authority issues private keys associated with a set of attributes $S \subseteq \mathcal{U}$ that describe the credentials or profile of the user — for example, {`Doctor, Cardiologist, Hospital-X, Senior-Level`} for a medical professional [3]. The private key contains no information about the access policy; it exclusively reflects the attributive identity of the user within that system.

The sender, in turn, specifies an access policy $\mathbb{A}$ — represented either as an access tree or, in modern constructions, as an LSSS matrix (M, ρ) — and associates it directly with the ciphertext in the encryption algorithm [13]. The message m is masked by multiplication with $e(g, g)^{\alpha s}$, where α is a secret parameter of the system and $s \in \mathbb{Z}_p$ is a random exponent chosen at each encryption. The secret s is distributed along the rows of the LSSS matrix according to the mechanism described in Section 2.2.2: the

vector $v = (s, r_2, \ldots, r_n)$ generates the shares $\lambda_i = (M \cdot v)_i$ for each row i, and each share is embedded into a ciphertext component associated with the attribute $\rho(i)$ [13]. The result is that the structure of the access policy is mathematically inscribed into the ciphertext, and recovering the message requires the reconstruction of s through the linear reconstruction property of the LSSS.

Decryption is possible if and only if the attribute set S from the user's private key satisfies the policy $\mathbb{A}$ in the ciphertext. In this case, the user identifies the subset $I = \{i : \rho(i) \in S\}$ of rows of M corresponding to their attributes and calculates the reconstruction constants $\{w_i\}_{i \in I}$ such that $\sum_{i \in I} w_i \lambda_i = s$. Applying pairing operations between the ciphertext components and the private key components for each $i \in I$, the user obtains terms of the form $e(g, g)^{\lambda_i \cdot f(t)}$, where t is the randomization factor of the key. Combining these terms with the coefficients w_i in the exponent cancels out both the shares λ_i and the factor t, recovering $e(g, g)^{\alpha s}$ and implicitly the message [13, 3].

Collusion prevention follows the same structural principle as in KP-ABE, but adapted to the new architecture. During the generation of each private key, the authority chooses a random exponent $t \in \mathbb{Z}_p$ independently for each user. All components of the key are tied to this t through the algebraic structure of the construction [3]. The consequence is that two users with attribute sets S_1 and S_2, each individually insufficient to satisfy the policy $\mathbb{A}$, cannot combine their key components: the factors $t_1 \neq t_2$ are algebraically incompatible, and the attempt to combine them produces residual terms that prevent the correct cancellation of the masking factor [13].

The importance of CP-ABE for cloud environments is direct and profound. A medical data owner can encrypt a file with the policy *"(Doctor AND Cardiologist) OR System-Administrator"* without knowing the specific identities of the users who satisfy these conditions and without interacting with them in advance [3, 15]. The cloud server storing the data cannot decrypt the file, because it does not hold the private key of any authorized user. Access control is thus achieved cryptographically, rather than through server-level access control policies, an essential property in the context of a threat model where the cloud server is considered semi-trusted (honest-but-curious).

3.4.2 Large Universe vs. Small Universe Constructions

A design decision with significant practical implications in any CP-ABE system concerns the way attributes are represented and managed at the level of system pa-

rameters. This choice determines the scalability of the scheme, the size of the public parameters, and the degree of flexibility in adding new attributes after the initialization phase.

In small universe constructions, the complete set of possible attributes must be exhaustively declared in the *Setup* phase [3]. The authority allocates an explicit mathematical element in the public parameters PK for each attribute in the universe — usually, a power of the generator g with an exponent derived from the identity of the attribute. The seminal construction of Bethencourt et al. [3] operates in this regime: the size of PK grows linearly with the cardinality of the attribute universe $|\mathcal{U}|$, which makes it unsuitable for systems with numerous or dynamic attributes. The advantage of this approach is the simplicity of the security proof and the efficiency of the decryption operations, because the necessary parameters are precomputed and directly available.

Large universe constructions eliminate this restriction, allowing the use of an exponentially large or practically infinite attribute space [13]. The central idea is that attributes are no longer enumerated in PK, but are derived on-the-fly through a hash function $H : \{0,1\}^* \to G$, which maps any arbitrary character string, for example, `"Allergic-Penicillin"` or `"Cardiology-Department-ER"`, directly into a group element [13]. This approach decouples the size of PK from the total number of attributes in the system: the public parameters now have a constant size or grow linearly with a parameter $n_{\max}$ representing the maximum number of attributes in a single access formula, not the total number of attributes in the universe [13, 10].

Waters' scheme [13], which forms the basis of most modern AB-PRE constructions, is a large universe construction that utilizes the q-Parallel BDHE assumption to achieve adaptive security in the standard model. The size of the public parameters grows linearly with $n_{\max}$, and the ciphertext size grows linearly with the number of attributes in the access formula, both essential properties for practical usability in cloud environments with complex policies [13]. By contrast, small universe constructions with adaptive security in the standard model, based on composite order groups in the spirit of Lewko and Waters' Dual System Encryption technique [10], achieve stronger guarantees, but at the price of a significantly higher computational overhead due to operations in composite order groups.

A particular case is worth mentioning: certain large universe constructions operating in the random oracle model manage to achieve constant-size public parameters,

independent even of $n_{\max}$, by programming the hash oracle in the security proof [7]. This offers maximum efficiency in terms of parameter size, but at the cost of the weaker guarantees associated with the random oracle model, discussed in Section 2.3.3.

3.5 Structural Comparison: KP-ABE vs. CP-ABE

The previous sections presented KP-ABE and CP-ABE as distinct paradigms, each with its own architecture. Viewed comparatively, however, the two constructions reveal a structural relationship that goes beyond simple analogy: they are consistently described in the literature as duals of each other, with the roles of keys and ciphertexts inverted [2, 3].

Role Inversion

The fundamental separation can be formulated concisely: in KP-ABE, the access structure $\mathcal{T}$ resides in the private key, and the attribute set γ resides in the ciphertext; in CP-ABE, the access structure resides in the ciphertext, and the attribute set S resides in the private key [2, 3]. Table 3.1 summarizes this inversion at the level of the main components of the two schemes.

Implications for Cloud Access Control

The two architectures reflect different philosophies regarding where the decision-making authority lies. In KP-ABE, the decision regarding who can access the data is made prior to encryption by the authority issuing the keys; the sender merely tags the data with descriptive attributes, without controlling the access policy [2]. This architecture is suitable for scenarios with stable, predefined roles (audit logs, broadcast with fixed categories), but becomes restrictive when the access policy must vary dynamically, depending on the sensitivity of each resource.

CP-ABE transfers this decision-making authority to the sender: the data owner defines the access policy at the moment of encryption, independently of the keys already issued, and can adapt it for each resource without interacting with the authority or the recipients [3, 13]. This property makes CP-ABE the dominant paradigm for cloud data sharing, where the owner of a resource stored on a semi-trusted server desires granular control over access, a scenario that KP-ABE cannot efficiently express.

Table 3.1: Structural comparison between KP-ABE and CP-ABE

Component	**KP-ABE [2]**	**CP-ABE [3]**
Private key contains	Access structure ($\mathcal{T}$)	Attribute set ($S \subseteq \mathcal{U}$)
Ciphertext contains	Attribute set ($\gamma \subseteq \mathcal{U}$)	Access structure ($\mathcal{T}$)
Secret sharing executed in	*KeyGen*	*Encrypt*
Secret shared at the tree root	(y) (Setup parameter)	(s) (ephemeral exponent, per encryption)
Decryption condition	($\mathcal{T}(\gamma) = 1$)	($S \models \mathcal{T}$)
Policy control	Central authority	Sender / data owner
Typical security assumption	BDH, selective model	GGM (Bethencourt et al.) / (q)-Parallel BDHE (Waters)
Natural use case	Audit logs, directed broadcast	Cloud data sharing, granular access

Asymmetry in Security Proofs

The structural symmetry breaks down at the level of security proofs. In KP-ABE, selective security reduces directly to BDH: the adversary declares the target attribute set γ^* before seeing the public parameters, allowing the simulator to apply a simple partitioning strategy [2].

In CP-ABE, the challenge ciphertext is associated with an access structure $\mathcal{T}^*$ (or, in LSSS-based constructions, a matrix M^*), in which the same attribute can appear in multiple positions. The simulator must embed this structure into the public parameters without having sufficient degrees of freedom under a simple assumption like BDH or DBDH. This is the reason why the original construction of Bethencourt et al. [3] could only be proven in the Generic Group Model, and subsequent LSSS-based constructions [13] required the q-Parallel BDHE assumption to achieve security in the standard model, as discussed in Section 2.1.2.

Synthesis and Transition

Viewed together, KP-ABE and CP-ABE represent two instantiations of the same idea: access control through the evaluation of a logical structure against a set of at-

tributes, differing in where the logical structure is placed. The question that this duality naturally raises is whether the inversion of roles can be made explicit at the level of the algebraic expressions that define the algorithms of the two schemes: can a transformation be identified which, applied to the KP-ABE construction of Goyal et al. [2], produces the CP-ABE construction of Bethencourt et al. [3]? And, if so, what part of the CP-ABE architecture results necessarily from this transformation, and what part represents an additional, independently introduced component?

This question constitutes the core of the original contribution of the present work and is formally addressed in Chapter 5, where the transformation $\mathcal{S}$ is defined and the algorithm-by-algorithm correspondence between the two constructions is demonstrated.

Chapter 4

Proxy Re-Encryption and the CP-ABPRE Construction

The previous chapters have progressively built the cryptographic foundations of attribute-based access control, culminating with CP-ABE as the dominant paradigm for granular data sharing in cloud environments. However, CP-ABE solves only half of the access delegation problem: it allows a data owner to specify who can decrypt a ciphertext, but it does not provide a mechanism by which decryption rights can be dynamically transferred from one user to another without the data being explicitly decrypted and re-encrypted. This capability, the cryptographic delegation of access without exposing the plaintext, is provided by an independent primitive with its own history and motivation: Proxy Re-Encryption (PRE).

PRE and ABE are paradigms that have evolved in parallel, each responding to distinct requirements. Their convergence produces AB-PRE constructions, cryptographic tools that combine the expressiveness of access policies in ABE with the flexibility of delegation in PRE. The most studied form in the literature is Ciphertext-Policy AB-PRE (CP-ABPRE), which this chapter analyzes in detail, alongside the dual KP-ABPRE construction, mentioned in the context of the structural relationship examined in Chapter 5. Understanding this convergence first requires a detailed analysis of PRE as an autonomous primitive, with its own structural properties, security models, and architectural tradeoffs.

4.1 Proxy Re-Encryption: Foundations and Syntax

The concept of proxy re-encryption was introduced by Blaze, Bleumer, and Strauss [5] under the name of atomic proxy cryptography, as a response to a concrete practical problem: how can an entity delegate the decryption capability of its messages to a third party, without revealing its private key to them and without having to manually decrypt and re-encrypt each message? The answer proposed by Blaze et al. is elegant in its conceptual simplicity: an intermediate entity is introduced — the proxy — which, holding a special re-encryption key, can transform a ciphertext intended for Alice into a semantically equivalent ciphertext, decryptable with Bob's private key, without obtaining any information about the plaintext message [5, 6].

Formally, a PRE scheme extends the syntax of a public key encryption system by adding two additional algorithms [6, 7]:

- **ReKeyGen**$(sk_A, pk_B) \rightarrow rk_{A\rightarrow B}$: Executed by Alice, using her own secret key sk_A and Bob's public key pk_B, it produces a re-encryption key $rk_{A\rightarrow B}$. This key can be entrusted to the proxy without compromising sk_A.

- **ReEnc**$(rk_{A\rightarrow B}, C_A) \rightarrow C_B$: Executed by the proxy, it transforms a ciphertext C_A encrypted under Alice's public key into a ciphertext C_B decryptable with Bob's secret key, or returns an error if the input is invalid.

An essential property that distinguishes PRE from simple manual re-encryption is that the proxy executes ReEnc without possessing sk_A and without being able to independently recover the plaintext from C_A or C_B [6]. The security of the scheme guarantees that the proxy, holding only $rk_{A\rightarrow B}$ and C_A, obtains no information about the underlying message, a guarantee formalized by the security models discussed in Section 2.3.

Many practical PRE constructions operate with two levels of ciphertexts [6, 15]. Level 2 ciphertexts are original ciphertexts, encrypted under Alice's public key, and can be input into the ReEnc algorithm for transformation. Level 1 ciphertexts are the ciphertexts resulting from re-encryption, decryptable directly with sk_B, but which can no longer be further re-encrypted. This stratification reflects a deliberate separation of rights: level 2 corresponds to the owner's raw data, level 1 corresponds to active delegations.

In cloud environments, the practical utility of PRE is direct. A data owner can

store level 2 ciphertexts on a cloud server and can issue selective re-encryption keys for different recipients, allowing the server to perform the necessary transformations on demand, without ever accessing the plaintext data [15, 19]. Access revocation reduces to deleting the corresponding re-encryption key, without requiring the re-encryption of the stored data.

4.1.1 Bidirectional vs. Unidirectional PRE

The first structural dimension that differentiates PRE constructions concerns the directionality of delegation: whether a re-encryption key allows transformation exclusively in the direction $A \to B$, or whether the inverse transformation $B \to A$ can also be deduced from it.

The original BBS98 scheme [5] is bidirectional: the re-encryption key has the form $rk_{A\to B} = sk_B/sk_A$ in the corresponding multiplicative group, which means that the multiplicative inverse $(rk_{A\to B})^{-1} = sk_A/sk_B$ implicitly constitutes the re-encryption key in the reverse direction [5, 6]. A proxy holding $rk_{A\to B}$ can, therefore, also transform ciphertexts intended for Bob into ciphertexts for Alice, even if Alice did not authorize this direction. Bidirectionality implies a symmetric trust relationship between the delegator and the delegatee, an unrealistic assumption in typical access control scenarios, where Alice wishes to grant access to Bob without receiving access to his data in return [6].

Unidirectional PRE eliminates this forced symmetry. Dodis and Ivan [26] first identified this property as a distinct design requirement, and Ateniese et al. [6] proposed the first efficient unidirectional construction based on pairings, the well-known AFGH scheme. In this construction, the re-encryption key $rk_{A\to B}$ is generated by combining sk_A with pk_B in a way that does not allow the deduction of $rk_{B\to A}$ without knowledge of sk_B [6]. The asymmetry is mathematically guaranteed through the use of bilinear maps: re-encryption involves a pairing operation that transforms a ciphertext component from the group G to the group G_T, and inverting this transformation is computationally infeasible under standard hardness assumptions [6, 7].

From the perspective of cloud applications, unidirectionality is the correct property for any asymmetric access delegation scenario. A system administrator granting temporary access to an external auditor does not want this to imply reciprocity. All relevant AB-PRE constructions in modern literature adopt unidirectional PRE as an

architectural foundation [7, 15, 19].

4.1.2 Single-Hop vs. Multi-Hop Properties

The second structural dimension concerns the number of successive transformations that a ciphertext can support: whether the re-encryption can be applied only once (single-hop) or an arbitrary number of times along a delegation chain (multi-hop).

In single-hop constructions, a level 2 ciphertext can be re-encrypted only once, producing a level 1 ciphertext that can no longer be input into the ReEnc algorithm [6]. The limitation is often a structural consequence of using pairings: the re-encryption operation applies the bilinear map $e : G \times G \to G_T$ to transform a ciphertext component from the base group G to the target group G_T. Because there is no standard bilinear map defined on $G_T \times G_T \to G_{T'}$ for another group $G_{T'}$, the chain of transformations naturally stops after the first step [6, 15]. This limitation is, in many contexts, a desirable security property: it restricts the proxy's capability to propagate unauthorized delegations.

Multi-hop constructions allow for repeated re-encryption, such that a ciphertext intended for Alice can be successively transformed for Bob, then for Carol, and so on [5]. Although useful in complex workflows with delegation chains, the multi-hop property introduces a specific risk: unwanted transitivity. In a multi-hop scheme based on the BBS pattern, a proxy holding the keys $rk_{A\to B}$ and $rk_{B\to C}$ can combine them to derive $rk_{A\to C}$, transferring access from Alice to Carol without Alice's consent [5, 7]. This property, termed unwanted transitivity in the specialized literature, undermines the data owner's control over the delegation chain and is considered a vulnerability in any system where Alice wishes to maintain ultimate authority over the redistribution of access [7].

Recent research aims to achieve the property of non-transferability, whereby combining the re-encryption key $rk_{A\to B}$ with the delegatee's secret key sk_B does not allow the generation of a new re-encryption key to a third party [19]. This property is distinct from collusion resistance and addresses a different attack vector: not collusion for decryption, but collusion for the unauthorized propagation of delegations.

4.1.3 Proxy Trust Model: Honest-but-Curious and Semi-Honest

Any PRE construction must be accompanied by an explicit specification of the trust model adopted for the proxy, because the security guarantees of the scheme are defined relative to the assumed behavior of this entity.

The standard model in the PRE and CP-ABPRE literature is that of the honest-but-curious proxy (also known as semi-honest) [6, 15]. An honest-but-curious proxy correctly executes all protocol algorithms, transforms ciphertexts according to the received re-encryption keys, deletes revoked keys upon request, but simultaneously attempts to extract any possible information from the data to which it has access: ciphertexts, re-encryption keys, and transformation results [15]. In the cloud context, the service provider naturally fits this model: it has economic incentives to honor the service agreement, but there are no technical guarantees that it will not access user data if it could do so without detectable consequences.

The central guarantee that a PRE scheme offers against an honest-but-curious proxy is that holding $rk_{A\to B}$ and any transformed ciphertexts does not allow the proxy to recover the plaintext or Alice's secret key, as long as it does not actively collude with Bob [6]. This guarantee is formalized in the IND-CPA or IND-CCA security game described in Section 2.3.1, extended with the specific PRE oracles mentioned there.

A distinct and more serious problem is collusion resistance against the collusion between the proxy and the delegatee. Early schemes, including BBS98 [5], suffered from a critical vulnerability: a proxy colluding with Bob could calculate $rk_{A\to B} \cdot sk_B$ to recover sk_A, thus obtaining total access to all of Alice's data [5, 6]. This property, termed master key security or collusion-safeness, requires that no combination of $rk_{A\to B}$ with sk_B allows the reconstruction of sk_A [6]. The AFGH scheme [6] solved this vulnerability by introducing an additional randomization factor into the re-encryption key generation, algebraically separating the components of the two keys and making the reconstruction of sk_A computationally equivalent to solving the DBDH problem.

The honest-but-curious model, combined with the master key security requirement, defines the minimum acceptable security framework for any PRE construction intended for cloud environments. The CP-ABPRE constructions analyzed in the following section invariably adopt this model as a starting point, adding on top the supplementary guarantees imposed by the integration with CP-ABE access policies.

4.2 CP-ABPRE: Combining ABE and PRE

The previous section presented PRE as an independent primitive, capable of delegating decryption rights between pairs of users identified by individual public keys. This form of delegation, a one-to-one mapping between identities, is sufficient for simple access redirection scenarios, but becomes inadequate in cloud environments where access rights are defined through complex logical policies, not individual identities. A medical data owner does not want to delegate access to Bob specifically, but to all users who satisfy the policy *"Doctor AND Cardiologist AND Hospital-X"*, a requirement that exceeds the expressive capacity of traditional PRE. The answer to this limitation is the fusion of CP-ABE and PRE into a unified primitive: Ciphertext-Policy Attribute-Based Proxy Re-Encryption (CP-ABPRE).

4.2.1 System Architecture and Syntax

The central intuition of CP-ABPRE is a generalization of re-encryption semantics: instead of transforming a ciphertext intended for an individual identity into a ciphertext for another identity, the proxy transforms a ciphertext associated with an access policy M into a ciphertext associated with a different access policy M', without accessing the plaintext [7, 15]. The validity condition of this transformation is no longer holding a re-encryption key generated from the pair (sk_A, pk_B), but satisfying the initial policy M by the attribute set S of the user issuing the delegation. In other words, only a user authorized to decrypt the original ciphertext can generate a valid re-encryption key for it, a property that cryptographically anchors the delegation in existing access rights.

The complete syntax of a standard CP-ABPRE scheme extends the basic algorithms of CP-ABE with two additional algorithms and refines the decryption algorithm to handle re-encrypted ciphertexts [7, 15, 19]. Formally, a CP-ABPRE scheme consists of the following algorithms:

- **Setup**$(\lambda) \rightarrow (PP, MSK)$: The initialization algorithm, executed by the central authority, takes the security parameter λ and produces the public parameters PP and the master secret key MSK. The structure of PP follows the conventions of large universe CP-ABE constructions, including the group generator g, the element $e(g,g)^{\alpha}$, and the family of terms necessary for the q-Parallel BDHE

assumption [13, 15].

- **KeyGen**$(MSK, S) \rightarrow sk_S$: The authority generates a private key sk_S for a user with the attribute set $S \subseteq \mathcal{U}$, applying individual randomization through the factor $t \in \mathbb{Z}_p$ specific to each issuance, guaranteeing collusion resistance according to the mechanism described in Section 3.4.1 [13].

- **Encrypt**$(PP, M, m) \rightarrow C_M$: Executed by the data owner, encrypts the message m against the access policy (M, ρ) represented as an LSSS matrix, producing a level 2 ciphertext C_M. The message is masked by $e(g, g)^{\alpha s}$, and the secret s is distributed through the LSSS mechanism along the ciphertext components, one per row of the matrix M [13, 15].

- **ReKeyGen**$(sk_S, S, M') \rightarrow rk_{S \rightarrow M'}$: The central delegation algorithm, executed by the data owner or an authorized user. It takes the private key sk_S, the attribute set S, and the new target access policy M', producing a re-encryption key $rk_{S \rightarrow M'}$. The key $rk_{S \rightarrow M'}$ implicitly embeds the verification that S satisfies the original policy: if this condition is not met, the proxy's transformation will algebraically fail [7, 15]. The non-interactive property of this algorithm, the fact that the generation of $rk_{S \rightarrow M'}$ does not require the participation of the recipients or knowledge of their identities, is essential for practical usability in the cloud, where the delegatee can be offline at the time of actual access.

- **ReEnc**$(rk_{S \rightarrow M'}, C_M) \rightarrow C_{M'}$: Executed by the proxy (cloud server), takes the level 2 ciphertext C_M and the key $rk_{S \rightarrow M'}$. Using bilinear maps, the proxy algebraically evaluates whether the attributes S embedded in the key satisfy the policy M from the ciphertext. If so, pairing operations cancel the component associated with the policy M and replace it with the components corresponding to the policy M', producing a level 1 ciphertext $C_{M'}$ [7, 15]. The proxy never recovers $e(g, g)^{\alpha s}$ during this operation; the masking factor remains intact, protected by the algebraic structure of the construction.

- **Decrypt**$(sk_S, C_M) \rightarrow m$: The direct decryption of a level 2 ciphertext, executed by a user whose attribute set S satisfies the policy M, through the linear reconstruction of the secret s and the recovery of $e(g, g)^{\alpha s}$, according to the standard CP-ABE mechanism [13].

- **DecryptR**$(sk_{S'}, C_{M'}) \rightarrow m$: The decryption of a re-encrypted level 1 ciphertext, executed by a user whose attribute set S' satisfies the new policy M'. The algebraic structure of $C_{M'}$ differs from that of a level 2 ciphertext, requiring a distinct decryption algorithm that takes into account the transformation applied by the proxy [7, 15].

An important architectural extension was proposed by Kawai [11] under the name of Flexible CP-ABPRE, which addresses a practical limitation of the standard scheme: whenever the target policy M' changes, the data owner must execute *ReKeyGen* using sk_S, exposing the secret key in memory and consuming significant computational resources. Kawai proposes separating ReKeyGen into two stages: the user executes a URKG algorithm only once, which produces a universal re-encryption key urk_S, independent of any specific target policy. Subsequently, a specialized re-encryption key generation authority (Re-encryption Key Generator, RKG) composes the specific keys $rk_{S \rightarrow M'}$ on-the-fly from urk_S and the parameters of the policy M', without requiring the intervention or online presence of the data owner [11]. This separation outsources the computational cost of adaptive delegation, reducing the exposure of the user's secret key to a single initial event.

4.2.2 Delegation of Decryption Rights in Cloud Settings

The CP-ABPRE architecture described in the previous section maps directly onto the threat model and operational requirements of modern cloud environments. The entities of a CP-ABPRE system deployed in the cloud correspond to well-defined roles: the central authority manages the system parameters and issues user keys; the data owner encrypts the resources and issues re-encryption keys; the cloud server acts as a semi-trusted proxy; the final recipients decrypt the re-encrypted ciphertexts with their own keys [15].

The typical operational flow illustrates the concrete advantages of this architecture. A doctor who wishes to store patient records in the cloud encrypts them against the policy *"(Doctor AND Cardiologist) OR Administrator"* and uploads them. Later, if a group of researchers with the attributes *"Researcher AND University-X"* requires temporary access to the same data, the doctor issues a re-encryption key $rk_{S \rightarrow M'}$ where M' corresponds to the researchers' group policy and transmits it to the cloud server. The server re-encrypts the records on demand, without accessing their content, and the re-

searchers can decrypt them with their own keys [7, 15]. The data owner does not need to be online at the time of access, does not download or re-encrypt any files, and access revocation reduces to deleting the re-encryption key from the server.

This property, termed offline delegation in the literature, is essential for mobility and IoT scenarios, where user devices have limited computational resources and intermittent connectivity [19]. The actual re-encryption, which involves multiple computationally expensive pairing operations, is fully outsourced to the cloud server, which has adequate processing power. The mobile user only transfers the re-encryption key $rk_{S \to M'}$, whose generation cost is comparable to a standard key generation operation, not to a full encryption operation [11, 19].

An additional capability of CP-ABPRE with direct implications for dynamic access administration is the extension and restriction of policies without re-encrypting the original data [7]. By issuing re-encryption keys with different target policies for the same resource, the owner can simultaneously create isolated access groups with distinct policies, all derived from the same stored ciphertext. The cloud server manages the transformations corresponding to each active re-encryption key, producing different level 1 ciphertexts for different groups, without the groups being able to interfere with each other or access the ciphertexts intended for other groups [15].

These properties have motivated proposals for the integration of CP-ABPRE into critical infrastructures beyond conventional cloud storage. In the context of smart grids, for example, intermediate nodes can act as proxies that transform the encrypted energy consumption logs of domestic users into formats accessible to analysts with the appropriate attributes, complying with data protection regulations without exposing the raw data [19]. The Proxy Re-Encryption as a Service (PREaaS) model, in which re-encryption operations are offered as a standardized cloud service, follows the same logic and represents an active direction of research in the field.

4.3 Security and Efficiency in CP-ABPRE

The fusion of CP-ABE and PRE produces a cryptographic tool with remarkable properties, but simultaneously introduces security and efficiency challenges that do not appear in either of the two systems taken separately. The presence of the proxy as an active entity in the ciphertext transformation process creates new attack vectors, and the expressiveness requirements of LSSS policies amplify the computational cost

of re-encryption operations. This section analyzes the three critical dimensions of security and efficiency in CP-ABPRE: collusion resistance and master key security, the difficulty of achieving CCA security in the presence of a proxy, and the practical trade-offs among parameter size, computational cost, and outsourcing possibilities.

4.3.1 Collusion Resistance and Master Key Security

Collusion resistance in CP-ABPRE is significantly more complex than in pure CP-ABE, because the presence of the re-encryption key $rk_{S\to M'}$ introduces an additional channel through which an adversary might attempt to reconstruct the delegator's long-term secrets. The relevant property, termed master key security and formally introduced for PRE schemes by Ateniese et al. [6], requires that no combination of $rk_{S\to M'}$ with the secret key $sk_{S'}$ of a delegatee whose attributes satisfy M' allows the recovery of sk_S or the decryption of ciphertexts for which the delegator has not explicitly issued a re-encryption key [6, 15].

The concrete attack scenario is the following: the proxy holds $rk_{S\to M'}$ and colludes with a delegatee who holds $sk_{S'}$ with $S' \models M'$. The goal of the collusion is either the extraction of sk_S, or obtaining the capability to decrypt ciphertexts encrypted against policies completely different from M, for which no re-encryption key has been issued [7, 15]. If the scheme is not explicitly designed against this attack, the algebraic components of $rk_{S\to M'}$ and $sk_{S'}$ could combine to cancel the masking factors and expose the secret parameters of sk_S.

The protection mechanism adopted in modern CP-ABPRE constructions consists of introducing additional randomization factors in the ReKeyGen algorithm, which mask the components of sk_S in the structure of the re-encryption key in a way that cannot be inverted without knowledge of additional secret values [7, 15]. Technically, this is achieved by adding random exponents $\theta_1, \theta_2 \in \mathbb{Z}_p$ chosen independently at each execution of ReKeyGen, such that the components of $rk_{S\to M'}$ are blinded relative to the values in sk_S through a randomization layer invisible to both the proxy and the delegatee [15]. During the ReEnc operation, the pairing operations evaluate the logic of the policy M and produce the necessary transformation, but the factors θ_1, θ_2 cancel each other out only in the specific context of the re-encrypted message, without ever isolating the individual components of sk_S in a recoverable form [7].

A complementary perspective on collusion in CP-ABPRE concerns the possibility

that two users with attribute sets S_1 and S_2, each individually insufficient to satisfy the policy M of a ciphertext, combine their re-encryption keys to generate a valid key. This form of collusion is prevented by the same individual per-user randomization mechanism present in CP-ABE: the factors $t_1 \neq t_2$ from the keys sk_{S_1} and sk_{S_2} make the components of the two re-encryption keys algebraically incompatible, preventing their coherent combination [13, 15].

4.3.2 The Challenge of CCA Security in PRE

Achieving IND-CCA security in a CP-ABPRE scheme represents one of the most difficult technical challenges in the field, and understanding the source of this difficulty requires returning to the fundamental tension identified in Section 2.3.1: the incompatibility between the rigid integrity required by CCA security and the proxy's transformation functionality.

The CHK transformation [20], the standard method for converting CPA schemes into CCA schemes in classical asymmetric cryptography, attaches a one-time signature (OTS) to the ciphertext to guarantee that it cannot be modified without invalidating the signature. Any attempt by an adversary to construct a modified ciphertext and send it to the decryption oracle is blocked by the signature verification. Applied naively to a CP-ABPRE system, this transformation becomes self-destructive: the proxy must structurally modify the ciphertext during the re-encryption process, and the OTS signature attached by the sender covers exactly the components that the proxy must transform [7, 15]. The result is that either the legitimate re-encryption invalidates the signature and blocks decryption, or the signature does not cover the components modified by the proxy and the scheme remains vulnerable to malicious modifications.

The solution adopted by Liang et al. [15] to achieve IND-sAS-CCA (Selective Access Structure and Chosen-Ciphertext Attack) security consists of a hybrid architecture that separates the ciphertext components based on their role in the re-encryption process. The sender signs with an OTS key the original ciphertext components that will *not* be modified by the proxy — in particular, the description of the LSSS access structure and the components associated with the policy M — and includes the OTS verification key in the ciphertext [15]. The proxy verifies the signature before executing the re-encryption: a ciphertext with an invalid signature is rejected, blocking modification attacks at the input level. The actual re-encryption transforms the algebraic compo-

nents associated with the policy M and produces a level 1 ciphertext that embeds the policy M', but no longer contains the original signed components; by construction, the re-encryption produces a new algebraic structure, not a modification of the old one [7, 15].

The re-encrypted ciphertext $C_{M'}$ has a stratified structure: an asymmetric component σ_2 contains the message masked against the policy M' in the form of a standard CP-ABE construction, and a symmetric component σ_1 encapsulates a derived session key, symmetrically encrypted with a CCA-secure system [15]. The final delegatee first extracts the session key from σ_2 through standard CP-ABE decryption, verifies a mathematical link embedded by the proxy to detect any modification of the symmetric component, and finally recovers the message from σ_1. This stratification guarantees that an adversary who modifies $C_{M'}$ after re-encryption cannot obtain useful information from the behavior of the decryption algorithm, because any modification is detected through the mathematical link verified by the delegatee [15].

The single-hop property is maintained by the deliberate exclusion of a structural component from $C_{M'}$: the re-encrypted ciphertext does not contain the factor g^s present in the original level 2 ciphertexts, a factor which is essential for the ReEnc algorithm to function [7]. In the absence of g^s, a new call to ReEnc with $C_{M'}$ as input fails algebraically, preventing the propagation of the re-encryption chain beyond a single step and maintaining the data owner's control over the granularity of delegation.

The scheme of Liang et al. [15] thus represents the first viable solution to the CCA security problem in CP-ABPRE, demonstrating that the tension between ciphertext integrity and proxy functionality can be resolved through the architectural separation of signed components from transformed ones. However, the security proof of this scheme operates in the random oracle model and under selective security—the adversary must commit to the target access structure before seeing the public parameters—two restrictions that limit the strength of the guarantees provided relative to the requirements of cloud scenarios with adaptive adversaries.

The current pinnacle of CCA security in CP-ABPRE is represented by the subsequent construction of Liang et al. [19], which eliminates both restrictions: the scheme achieves adaptive CCA security directly in the standard model, without the idealization of hash functions and without the early commitment to the target policy. This is achieved by adopting the Dual System Encryption technique on composite order bilinear groups $N = p_1p_2p_3$, which allows the indistinguishable transition between normal

and semi-functional keys and ciphertexts described in Section 2.3.2, now extended to the proxy re-encryption context [19, 10]. The price of these superior guarantees is a significantly higher computational and storage overhead compared to the 2014 construction, due to the operations in composite order groups, a direct trade-off between the strength of theoretical guarantees and practical efficiency, synthesized in the comparative analysis in Section 4.4.

The two constructions of Liang et al. together define the current frontier of CCA security in CP-ABPRE: [15] for scenarios where efficiency is a priority and the random oracle model is acceptable, and [19] for critical applications that require complete formal guarantees against adaptive adversaries in the standard model.

4.3.3 Ciphertext Size, Computational Overhead, and Outsourcing

The practical performance of CP-ABPRE schemes is determined by three main factors: the size of the ciphertexts, the cost of pairing operations, and the distribution of this cost among the system's entities.

The size of an original level 2 ciphertext in constructions based on Waters CP-ABE [13] grows linearly with the number of attributes in the access formula, i.e., $O(|M|)$ group elements, independent of the attribute universe size $|\mathcal{U}|$, an essential property for scalability in cloud environments with complex policies [13, 15]. The re-encrypted level 1 ciphertexts have a similar size, dominated by the components associated with the policy M', with a constant additional overhead for the symmetric components and integrity links [15].

The most computationally expensive operation remains ReEnc, executed by the proxy. The number of pairing operations required is proportional to the number of attributes in the set S that actually participate in satisfying the policy M, resulting in an asymptotic cost of $O(|S|)$ pairings [7, 15]. Because pairing operations are significantly more expensive than exponentiations in elliptic groups, with order-of-magnitude differences that vary depending on the implementation and hardware platform, outsourcing the ReEnc operation to the cloud server is essential for the usability of the scheme on resource-constrained devices [19, 11].

An additional practical limitation arises in the dynamic management of delegations: whenever the target policy M' changes, due to organizational restructuring, access revocation, or the addition of new user groups, the data owner must execute

ReKeyGen using sk_S, exposing the secret key in memory and consuming computational resources at each change [11]. In IoT or mobile scenarios, where user devices have limited processing capacity and intermittent connectivity, this requirement becomes a significant operational obstacle.

The Flexible CP-ABPRE mechanism described in Section 4.2.1 directly addresses this limitation by separating the computational cost of ReKeyGen between the user and the RKG authority: after the initial *URKG* phase, the operational complexity at the user level becomes constant, $O(1)$, regardless of the number or frequency of subsequent policy changes [11, 19]. This trade-off, an amortized fixed cost at the user level in exchange for fully outsourcing the adaptive key generation to the RKG, is representative of the general direction of efficiency optimization in CP-ABPRE: shifting the computational cost from resource-constrained devices to entities with dedicated processing power.

4.4 Comparative Analysis of Existing CP-ABPRE Schemes

The discussion in the previous sections has essentially followed a single evolutionary trajectory: that of security guarantees, from CPA to CCA, from the random oracle to the standard model, and from selective commitment to the adaptive adversary. However, the relevant comparative space for the field is not limited to this single dimension. Two other axes remain to be mapped, equally relevant for positioning CP-ABPRE constructions relative to one another: the degree of expressiveness of the access policy that each scheme can support—an axis that has developed relatively independently of the security one, and not infrequently in tension with it—and the directions through which the core functionality was subsequently extended, without modifying the central algebraic structure of the scheme, but rather by adding supplementary properties required by concrete use-case scenarios.

The Evolution of Access Policy Expressiveness

The construction by Liang et al. [8], which formally introduced the concept of attribute-based proxy re-encryption, operated with an expressiveness significantly inferior to that discussed in Sections 4.2 – 4.3. The access policy was reduced to a single conjunctive formula (AND-gate) over positive and negative attributes, without allowing the expression of disjunctions or threshold structures. The security guarantee ob-

tained, in the form of IND-SS-CPA, alongside the collision resistance demonstrated under the ADBDH assumption, nevertheless established the conceptual pattern that subsequent constructions would reiterate: the re-encryption key is derived from the private components of the delegator and is functional exclusively when their attribute set satisfies the initial policy, the same syntactic principle presented in Section 4.2.1 for the general formulation on LSSS.

A contemporary ramification of this construction, but divergent from a structural point of view, was proposed by Mizuno and Doi [12]: the first scheme capable of transferring the content of an ABE ciphertext into a form decryptable by a separate IBE scheme, with security proven in the standard model under the DBDH assumption. Unlike the dominant trajectory of CP-ABPRE, the proxy in this construction no longer produces a CP-ABE ciphertext governed by a new policy, but crosses the boundary between two heterogeneous cryptographic primitives, relinquishing the structural homogeneity of the output in favor of compatibility with already implemented IBE systems. The construction practically delineates the conceptual margin of what can be categorized as CP-ABPRE in the restrictive sense used here: a scheme in which both the input and the result of the ReEnc algorithm remain, without exception, CP-ABE ciphertexts.

Starting from the same AND-gate structure, Luo et al. [9] extend the scheme of Liang et al. [8] by introducing multi-value attributes and wildcards, but also through a feature absent from the original variant: the ability for the data owner to decide, right from the moment of encryption, whether a certain ciphertext can or cannot be subjected to the re-encryption operation by the proxy. The security reduction remains anchored in the selective model, under the DBDH assumption in its standard form, simpler than ADBDH, but beholden to the same limitation to attribute conjunctions, without opening the space towards more expressive structures.

The decisive step towards general expressiveness, the abandonment of AND-gate formulas in favor of arbitrary monotonic access structures, encoded by the matrix-function pair (M, ρ), is achieved by the constructions examined extensively in Section 4.3.2 [7, 15, 19], which adopt the LSSS convention introduced by Waters [13] and presented in Section 4.2.1, thus removing the conjunctive restriction without sacrificing any collusion resistance. The paper [7], initially presented at an IEEE conference in 2013, and its extension [15], published in 2015, fundamentally correspond to the same IND-sAS-CCA construction proven in the ROM, already analyzed in Section 4.3.2; the

paper [19] marks the subsequent step towards adaptive security achieved directly in the standard model. The synthesis of this evolution is presented in Table 4.1.

Table 4.1: Evolution of access policy expressiveness and security in early CP-ABPRE constructions

Scheme	Access Policy	Security	Model	Remarks
Liang et al. [8]	AND-gate (positive/negative attributes)	CPA, selective (ADBDH)	Standard	Initial reference framework of the field
Mizuno & Doi [12]	N/A (heterogeneous output)	CPA (DBDH)	Standard	ABE $\rightarrow$ IBE re-encryption; structural divergence from canonical CP-ABPRE
Luo et al. [9]	Extended AND-gate (wildcards, multi-value)	CPA, selective (DBDH)	ROM	Adds re-encryption control by the DO
Liang et al. [7, 15]	Monotonic LSSS (M, ρ)	CCA (IND-sAS-CCA), selective	ROM	Hybrid CHK-OTS architecture (cf. §4.3.2)
Liang et al. [19]	Monotonic LSSS (M, ρ)	CCA, adaptive	Standard	Dual System Encryption, composite order (cf. §4.3.2)

Reading the table horizontally confirms the observation already made in Sections 4.1.1 and 4.1.2: regardless of the generation they belong to or the expressiveness of the policy they support, all the listed constructions remain anchored in the unidirectional and single-hop paradigm, a limitation that does not stem from design choices, but from the algebraic constraints of the bilinear maps discussed in Section 4.1.2, constraints that no subsequent functional extension has managed to remove.

Directions of Functional Extension Post-2015

Once the security and expressiveness foundations of the basic architecture were stabilized, subsequent research turned towards extending functionality in directions that leave the algebraic core of the ReKeyGen and ReEnc algorithms unchanged, but which address practical limitations left open in the description from Section 4.2.1.

The first of these limitations concerns the privacy of the access policy itself. As formalized in Section 4.2.1, the structure (M, ρ) circulates unencrypted alongside the

ciphertext, thus exposing to the proxy, but also to any observer with access to the storage channel, information about the access conditions, even if the message content remains protected. Zhang et al. [27] respond to this leakage through a technique they term match-then-re-encrypt: the proxy evaluates whether an attribute set satisfies a hidden policy without needing the explicit reconstruction of the structure (M, ρ). The result is a notion of anonymity that complements message confidentiality, proven in the selective model under a combination of assumptions: DBDH, DLIN, and CBDH.

A second limitation concerns how user revocation interacts with the re-encryption mechanism. The ReKeyGen algorithm, as defined in Section 4.2.1, redirects access to a new policy, but does not provide any dedicated means for the direct exclusion of a single user without entirely regenerating the policy. Yang et al. [28] construct an access tree-based scheme, with constant-size attribute keys, in which policy updating through re-encryption and direct revocation coexist as separate operations; the particularity of this proposal is the security proof performed in the Generic Group Model, a choice that distances it from the reductions to standard computational assumptions used in Section 4.3.

The third direction addresses a transversal vulnerability in all the constructions discussed up to this point: the reliance on bilinear maps, structures compromised under quantum attacks. Li et al. [29] propose the first CP-ABPRE scheme built exclusively on hard lattice problems (LWE), retaining the ReKeyGen/ReEnc syntax from Section 4.2.1, but fully substituting the bilinear apparatus with trapdoor sampling and matrix computation over $\mathbb{Z}_q$. The security level achieved remains, for the time being, selective, below the adaptive guarantee offered by the pairing-based constructions in Section 4.3.2, which makes visible the trade-off between post-quantum resistance and the strength of the security proof.

The fourth direction concerns the assumptions regarding the proxy's behavior. The honest-but-curious model, established in Section 4.1.3, assumes a proxy that respects the protocol but seeks to extract information from what it has available. Ge et al. [30] relax this premise, proposing a construction in which the delegatee can verify, after receiving the re-encrypted ciphertext, the correctness of the applied transformation, thus identifying a proxy that deviates from the protocol, a behavior that the honest-but-curious model does not anticipate.

The fifth direction addresses an architectural risk left undiscussed: the dependence of the entire system on a single authority, responsible for issuing keys according

to the Setup/KeyGen in Section 4.2.1, whose compromise would affect the system as a whole. Yi et al. [31] describe a construction in which multiple attribute authorities, alongside an authorization center, jointly generate the system keys, thus eliminating the single point of failure; the scheme additionally introduces attribute weights, through which the policy becomes capable of expressing multiple values, not just binary ones, without increasing the number of distinct attributes in the system's universe.

Viewed together, the five directions—policy privacy, revocation integration, post-quantum resistance, re-encryption verifiability, and authority decentralization—share the same pattern: none of them intervenes in the algebraic relationship between the private components of the delegator's key and the re-encryption key $rk_{S \to M'}$, as established in Section 4.2.1. The discussed extensions therefore enrich the peripheral properties of the system, but leave open the question regarding the origin of that algebraic relationship, a question that Chapter 5 addresses directly through the structural analysis of the connection between KP-ABE and CP-ABE.

Chapter 5

A Structural Transformation from KP-ABE to CP-ABE

5.1 Motivation

The previous chapters have followed, in a deliberately progressive order, the conceptual evolution from identity-based cryptography to the KP-ABE and CP-ABE paradigms (Sections 3.3 and 3.4), culminating in their synthesis within CP-ABPRE (Chapter 4). Throughout this trajectory, the relationship between KP-ABE and CP-ABE has been presented, consistent with the specialized literature, as an *architectural inversion*: what in KP-ABE is embedded in the private key (the access structure) becomes, in CP-ABE, part of the ciphertext, and what in KP-ABE labels the ciphertext (the attribute set) becomes, in CP-ABE, the content of the private key (Section 3.4.1).

This characterization, although correct at an intuitive level and sufficient to motivate the historical transition between the two paradigms, remains essentially descriptive. Even Goyal et al. [2], in the extensions and open problems section of their original paper, observe the possibility of a "reverse" scheme, in which keys are associated with attribute sets and ciphertexts with policies, without, however, providing an actual construction or an analysis of the relationship between the two architectures at the algorithmic level. This observation has remained, in subsequent literature, more of a programmatic note than a formal result: the CP-ABE construction proposed a few years later by Bethencourt, Sahai, and Waters [3] was developed independently, using a pair of Setup exponents (α, β) and a per-attribute randomization mechanism based on a random oracle H, components that do not appear in their explicit form in the

KP-ABE construction by Goyal et al. [2].

A natural observation that motivates this analysis concerns the overall architecture of the work: if the relationship between KP-ABE and CP-ABE can be made explicit at the algebraic level, then the trajectory KP-ABE $\xrightarrow{\mathcal{S}}$ CP-ABE $\rightarrow$ CP-ABPRE suggests that any CP-ABPRE scheme carries, in its algebraic substrate, an origin in the initial KP-ABE construction. Figure 1.1 illustrates this structure: the transformation $\mathcal{S}$ (arrow ①) and the fusion of CP-ABE with PRE (arrow ②) compose a path from KP-ABE to CP-ABPRE, simultaneously raising the symmetric question of whether an analogous path exists directly from KP-ABPRE to CP-ABPRE, a question treated as an open research direction in Section 5.4 and in Chapter 6.

The question that this chapter addresses is, therefore, a precise one: can a transformation be made explicit, at the level of the algebraic expressions defining $Setup$, $Encrypt$, $KeyGen$, and $Decrypt$, which, applied to the KP-ABE construction from [2], produces the CP-ABE construction from [3]? And, if so, what exactly from the structure of the resulting CP-ABE scheme is a forced consequence of this transformation, and what exactly represents an additional component, introduced independently?

The answer constructed in this chapter, the original contribution of the present dissertation, is affirmative and is formulated as a transformation $\mathcal{S}$, defined through the composition of three elementary operations applied to the algorithmic components of the KP-ABE scheme. We demonstrate that this transformation reproduces, modulo exactly three identifiable substitutions, the CP-ABE scheme of Bethencourt et al. [3]. Moreover, one of these three substitutions, the appearance of the random oracle H in place of the per-attribute master key components, is not an arbitrary choice, but a demonstrable consequence of the other two operations, under a locality constraint on the master key that we formalize explicitly in Section 5.3.4.

This analysis provides an answer to one of the research questions formulated in Chapter 1, regarding the structural relationship between the two ABE paradigms that underlie CP-ABPRE, and also constitutes the foundation for the discussion in Sections 5.3 and 5.4, where the implications of the transformation $\mathcal{S}$ are extended towards the characterization of the classes of CP-ABE schemes accessible through this path, respectively towards the role of the ReKeyGen algorithm in the CP-ABPRE architecture (Section 4.2.1) and towards the structural relationship between KP-ABPRE and CP-ABPRE as instantiations of the same AB-PRE primitive (Section 5.4).

We emphasize from the outset the methodological boundary of this analysis: the

transformation $\mathcal{S}$ is defined and analyzed exclusively at the level of the concrete algebraic expressions from [2] and [3], polynomials, exponents, bilinear pairings, without resorting to a higher-level formal apparatus, such as category theory or notions of morphism between cryptographic schemes. This choice is deliberate: the objective is a step-by-step verifiable proof, anchored in the original constructions, not an abstract reformulation of the relationship between them.

5.2 The Transformation $\mathcal{S}$: From KP-ABE to CP-ABE

5.2.1 Unified Notation and the Two Source Schemes

We harmonize the notation of the two works (which use G_0/G_1 and G_1/G_2, respectively, for the same roles) to the convention (G, G_T, e), $e : G \times G \to G_T$, generator $g \in G$, group of prime order p, established in Section 2.1.1.

Scheme Π_{KP} (KP-ABE Construction [2]):

$$
\begin{aligned}
Setup_{KP}: &\quad \text{for } i \in \mathcal{U} = \{1, \dots, n\} : t_i \xleftarrow{\$} \mathbb{Z}_p, \quad y \xleftarrow{\$} \mathbb{Z}_p, \\
&\quad PK_{KP} = \big(\{T_i = g^{t_i}\}_{i \in \mathcal{U}},\ Y = e(g,g)^y\big), \quad MK_{KP} = \big(\{t_i\}, y\big). \\
Encrypt_{KP}(M, \gamma): &\quad s \xleftarrow{\$} \mathbb{Z}_p, \quad CT = \big(\gamma,\ E' = M \cdot Y^s,\ \{E_i = T_i^s\}_{i \in \gamma}\big). \\
KeyGen_{KP}(\mathcal{T}): &\quad \text{polynomials } q_x \text{ top-down, } q_r(0) = y; \quad \forall x \in Leaves(\mathcal{T}),\ i = att(x) : \\
&\quad D_x = g^{q_x(0)/t_i}. \\
Decrypt_{KP}: &\quad e(D_x, E_i) = e(g,g)^{s \cdot q_x(0)} \xrightarrow{\text{Lagrange}} e(g,g)^{sy} = Y^s, \quad M = E'/Y^s.
\end{aligned}
$$

Scheme Π_{CP} (CP-ABE Construction [3]):

$$Setup_{CP}: \quad \alpha, \beta \xleftarrow{\$} \mathbb{Z}_p, \quad PK_{CP} = \big(g,\ h = g^{\beta},\ f = g^{1/\beta},\ e(g,g)^{\alpha}\big), \quad MK_{CP} = (\beta, g^{\alpha}).$$

$$\begin{aligned} Encrypt_{CP}(M, \mathcal{T}): \quad & s \xleftarrow{\$} \mathbb{Z}_p, \quad \text{polynomials } q_x \text{ top-down, } q_R(0) = s; \\ & CT = \Big(\mathcal{T},\ \tilde{C} = Me(g,g)^{\alpha s},\ C = h^s, \\ & \forall x \in Y:\ C_x = g^{q_x(0)},\ C'_x = H(att(x))^{q_x(0)}\Big). \end{aligned}$$

$$\begin{aligned} KeyGen_{CP}(S): \quad & r \xleftarrow{\$} \mathbb{Z}_p,\ \forall j \in S:\ r_j \xleftarrow{\$} \mathbb{Z}_p, \\ & SK = \Big(D = g^{(\alpha+r)/\beta},\ \forall j \in S:\ D_j = g^r H(j)^{r_j},\ D'_j = g^{r_j}\Big). \end{aligned}$$

$$\begin{aligned} Decrypt_{CP}: \quad & \frac{e(D_i, C_x)}{e(D'_i, C'_x)} = e(g,g)^{r \cdot q_x(0)} \xrightarrow{\text{Lagrange}} A = e(g,g)^{rs}, \\ & M = \tilde{C} \big/ \big(e(C,D)/A\big), \quad e(C,D) = e(g,g)^{(\alpha+r)s}. \end{aligned}$$

It can be immediately observed that the BSW scheme [3] explicitly uses the same access tree notation as [2] (the functions $att(x)$, $index(x)$, $parent(x)$, the Lagrange coefficients $\Delta_{i,S}$, the recursive $DecryptNode$ function). This notational coincidence is not accidental; it reflects an algorithmic identity that we formalize below.

5.3 The Secret-Sharing Layer $\mathcal{L}$

Definition 5.1 (Polynomial assignment $\mathcal{L}^{poly}$) *Let $\mathcal{T}$ be an access tree with root r (resp. R) and $\sigma \in \mathbb{Z}_p$. The procedure $\mathcal{L}^{poly}(\mathcal{T}, \sigma)$ recursively assigns, top-down, to each node $x \in \mathcal{T}$ a polynomial $q_x \in \mathbb{Z}_p[X]$ of degree $d_x = k_x - 1$, such that:*

- *$q_{root}(0) = \sigma$, the remaining d_{root} points chosen randomly;*
- *for $x \neq$ root: $q_x(0) = q_{parent(x)}(index(x))$, the remaining points chosen randomly.*

The output is the family $\{q_x(0)\}_{x \in Leaves(\mathcal{T})}$.

Lemma 5.2 (Identity of the sharing procedure) *The procedure used in $KeyGen_{KP}(\mathcal{T})$ to obtain $\{q_x(0)\}_{x \in Leaves(\mathcal{T})}$ (with $q_r(0) = y$) and the procedure used in $Encrypt_{CP}(M, \mathcal{T})$ to obtain $\{q_x(0)\}_{x \in Leaves(\mathcal{T})}$ (with $q_R(0) = s$) are the* same *procedure $\mathcal{L}^{poly}$, applied with different arguments:*

$$KeyGen_{KP}.\{q_x(0)\} = \mathcal{L}^{poly}(\mathcal{T}, y), \qquad Encrypt_{CP}.\{q_x(0)\} = \mathcal{L}^{poly}(\mathcal{T}, s).$$

Proof 1 *Direct comparison of the two descriptions (the* Key Generation *paragraphs in [2] and* Encrypt *in [3]): both specify exactly the same steps, setting the degree $d_x = k_x - 1$ for each node, fixing the value $q_{root}(0)$ equal to the secret relevant to the respective scheme (y, respectively s), propagating $q_x(0) = q_{parent(x)}(index(x))$ for the interior nodes, and randomly completing the other points of each polynomial. The only difference is the name of the value fixed at the root (y, respectively s) and the algorithm in which the procedure is invoked ($KeyGen$, respectively $Encrypt$).*

The $\mathcal{L}^{poly}$ procedure produces only scalar values $q_x(0) \in \mathbb{Z}_p$. The transformation of these scalars into group elements, the associated group component, differs essentially between the two schemes, and this difference is the point where the random oracle H appears.

Definition 5.3 (Group component $\mathcal{L}^{grp}$) *For a leaf x with $i = att(x)$ and the value $q_x(0)$ produced by $\mathcal{L}^{poly}$, the associated group component is:*

$$\mathcal{L}^{grp}_{KP}(x) = g^{\,q_x(0)/t_i} \quad (\text{uses } t_i \in MK_{KP}), \qquad \mathcal{L}^{grp}_{CP}(x) = \left(g^{\,q_x(0)},\ H(i)^{\,q_x(0)}\right) \quad (\text{uses only } H,\ PK_{CP}).$$

The essential difference is the following: $\mathcal{L}^{grp}_{KP}$ requires MK_{KP} (through t_i), whereas $\mathcal{L}^{grp}_{CP}$ requires no secret; it uses only the public hash function H. We will show in §5.3.4 that this difference is forced by the location where each group component must be produced.

5.3.1 The Per-Attribute Randomization Component $\mathcal{F}$ and the Recombination Mechanism $Combine$

Definition 5.4 (Per-attribute randomness component $\mathcal{F}$) *For a set of attributes X and a fresh scalar $c \in \mathbb{Z}_p$:*

$$\mathcal{F}_{KP}(\gamma, s) = \{E_i = g^{s \cdot t_i}\}_{i \in \gamma} \qquad (\text{uses } T_i = g^{t_i} \in PK_{KP}),$$

$$\mathcal{F}_{CP}(S, r) = \{(D_j, D'_j) = (g^r \cdot H(j)^{r_j},\ g^{r_j})\}_{j \in S} \qquad (\text{uses } H,\ r_j \xleftarrow{\$} \mathbb{Z}_p).$$

Definition 5.5 (Recombination mechanism $Combine$) *Given a tree $\mathcal{T}$ satisfied by the attribute set X, a family $\{q_x(0)\}_{x \in Leaves(\mathcal{T})} = \mathcal{L}^{poly}(\mathcal{T}, \sigma)$ and a scalar c, $Combine$ computes for each satisfied leaf x (with $i = att(x) \in X$) the value $e(g,g)^{c \cdot q_x(0)}$ through a bilinear pairing evaluation between $\mathcal{L}^{grp}(x)$ and the corresponding component from $\mathcal{F}(X, c)$, then recursively combines, using $F_x = \prod_{z \in S_x} F_z^{\Delta_{i, S'_x}(0)}$ (Lagrange interpolation), to obtain at the root*

$$Combine\big(\mathcal{L}^{poly}(\mathcal{T}, \sigma), \mathcal{F}(X, c)\big) = e(g,g)^{c \cdot \sigma}.$$

Lemma 5.6 (Common reduction to $e(g,g)^{c\sigma}$)

$$\textit{In } \Pi_{KP}: \quad e(D_x, E_i) = e\big(g^{q_x(0)/t_i}, g^{st_i}\big) = e(g,g)^{s\cdot q_x(0)} \xrightarrow{Combine} e(g,g)^{s\cdot y} \quad (c=s,\ \sigma=y).$$

$$\textit{In } \Pi_{CP}: \quad \frac{e(D_i, C_x)}{e(D_i', C_x')} = e(g,g)^{r\cdot q_x(0)} \xrightarrow{Combine} e(g,g)^{r\cdot s} \quad (c=r,\ \sigma=s).$$

Proof 2 *The first identity is the direct calculation from [2],* Decrypt. *The second is the direct calculation from [3],* Decrypt; *using $e(H(i),g)^{r_i q_x(0)} = e(g,H(i))^{r_i q_x(0)}$ (the symmetry of the bilinear pairing), the terms with $H(i)$ cancel out exactly. Applying Lemma 5.2 yields $\sigma \in \{y, s\}$ as stated.*

5.3.2 The Dual Role of $Combine$'s Output

Lemma 5.6 shows that $Combine$ produces, in both schemes, an expression of the form $e(g,g)^{c\sigma}$. However, the way this expression is subsequently used differs, and this difference constitutes the core of the transformation.

Proposition 5.7 (Masking function vs. correction function)

1. *In Π_{KP}, $\sigma = y \in MK_{KP}$ is a* Setup *secret, and $Combine$ produces* directly *the message masking factor: $e(g,g)^{sy} = Y^s$, used as $E' = M \cdot Y^s$.*

2. *In Π_{CP}, $\sigma = s$ is a* fresh Encrypt *variable and $c = r$ is a* fresh KeyGen *variable; neither is a Setup secret, therefore $Combine$ cannot directly produce a masking factor tied to MK_{CP}. The masking factor is $e(g,g)^{\alpha s}$ (with $\alpha \in MK_{CP}$), obtained from a* separate *pairing $e(C,D) = e(g,g)^{(\alpha+r)s} = e(g,g)^{\alpha s} \cdot e(g,g)^{rs}$. The output of $Combine$, $e(g,g)^{rs}$, plays the role of a* correction term, *which is removed from $e(C,D)$ to isolate $e(g,g)^{\alpha s}$.*

Proof 3 *(1) immediately from $E' = MY^s$ and Lemma 5.6. (2) immediately from $\tilde{C}/(e(C,D)/A) = M$ with $A = e(g,g)^{rs}$, according to the calculation in §4.1, substituting $e(C,D) = e(g,g)^{(\alpha+r)s}$ we obtain $\tilde{C} \cdot A/e(C,D) = Me(g,g)^{\alpha s} \cdot e(g,g)^{rs}/(e(g,g)^{\alpha s}e(g,g)^{rs}) = M$.*

Proposition 5.7 identifies, in exact algebraic terms, what happened to the role of the "secret shared on the tree": in KP-ABE this role coincides with the role of the *message masking exponent*; in CP-ABE the two roles are *separated*, the first being played by s (fresh, shared at Encrypt), and the second by α (fixed, from Setup, accessed through the pairing $e(C,D)$).

5.3.3 The Transformation $\mathcal{S}$

We define $\mathcal{S}$ as a composition of three operations, each motivated by the observations above.

Definition 5.8 (Transformation $\mathcal{S}$) *Starting from $\Pi_{KP} = (Setup_{KP}, Encrypt_{KP}, KeyGen_{KP}, Decrypt_{KP})$, $\mathcal{S}(\Pi_{KP})$ is obtained as follows:*

(T1) Translating the sharing procedure $\mathcal{L}^{poly}$. *The procedure $\mathcal{L}^{poly}(\mathcal{T}, \cdot)$ moves from $KeyGen$ to $Encrypt$. The tree $\mathcal{T}$ becomes part of the ciphertext; the value fixed at the root is no longer the Setup secret y, but a fresh random variable s, generated at each $Encrypt$.*

(T2) Translating the per-attribute randomness component $\mathcal{F}$. *The component $\mathcal{F}$ moves from $Encrypt$ to $KeyGen$. The attribute set (denoted γ in Π_{KP} and S in Π_{CP}) becomes part of the private key; the common exponent is no longer s, but a fresh random variable r, generated at each $KeyGen$.*

(T3) Recomposing the message masking mechanism. *The Setup secret y is renamed α and retains its role as the message masking exponent, but since $\mathcal{L}^{poly}$ no longer shares α (according to T1, it shares s), access to $e(g,g)^{\alpha s}$ is no longer direct. A new global Setup parameter β and a pair $(C, D) = (g^{\beta s}, g^{(\alpha+r)/\beta})$ are introduced such that $e(C, D) = e(g,g)^{(\alpha+r)s}$; the output of $Combine$ from $\mathcal{F}(S, r)$/$\mathcal{L}(\mathcal{T}, s)$, namely $e(g,g)^{rs}$ (Lemma 5.6), is used as a correction term to isolate $e(g,g)^{\alpha s}$ (Proposition 5.7, case 2).*

5.3.4 Necessity of the Random Oracle H

According to T1, the group component $\mathcal{L}^{grp}$ must be produced within the $Encrypt$ algorithm. But $\mathcal{L}^{grp}_{KP}(x) = g^{q_x(0)/t_i}$ requires $t_i \in MK_{KP}$, a Setup secret to which $Encrypt$ does not have access (by definition, $Encrypt$ only receives PK).

Proposition 5.9 (Impossibility of transporting the group component $\mathcal{L}^{grp}_{KP}$ under T1) *There is no group component $\mathcal{L}^{grp}$ that, taking as input PK_{KP} (without MK_{KP}), can produce $g^{q_x(0)/t_i}$ for an arbitrary $q_x(0)$, given that $T_i = g^{t_i}$ is the only available information about t_i; this would be equivalent to solving the discrete logarithm problem (DLP) in G.*

Proof 4 *The calculation of $g^{q_x(0)/t_i}$ from $q_x(0) \in \mathbb{Z}_p$ and $T_i = g^{t_i} \in G$ requires either t_i in the clear (thus MK_{KP}), or the extraction of t_i from T_i (discrete logarithm in G), which is infeasible under the standard group assumption.*

Therefore, under T1, the group component $\mathcal{L}^{grp}$ must be replaced with a component that requires no per-attribute secret, the solution being $\mathcal{L}^{grp}_{CP}(x) = (g^{q_x(0)}, H(i)^{q_x(0)})$, using the public H. Symmetrically, T2 moves $\mathcal{F}$ to $KeyGen$ (where $MK_{CP} = (\beta, g^{\alpha})$ contains no per-attribute components, in contrast to MK_{KP}); so that $Combine$ still produces $e(g,g)^{r \cdot q_x(0)}$ with the *same* r for all leaves (a necessary condition for Lagrange interpolation, see the step $F_x = \prod_z F_z^{\Delta(0)}$), a per-attribute cancellation mechanism is necessary, introduced for this purpose at $KeyGen$, based on fresh random values r_j tied to $H(j)$: the pair $(D_j, D'_j) = (g^r H(j)^{r_j}, g^{r_j})$, which cancels out exactly against $(C_x, C'_x) = (g^{q_x(0)}, H(i)^{q_x(0)})$ produced by $\mathcal{L}^{grp}_{CP}$.

Proposition 5.9 establishes, therefore, a precise impossibility: the *direct* transport of $\mathcal{L}^{grp}_{KP}$ under (T1) is excluded by the locality constraint of the master key. What the proposition does not establish is the uniqueness of the replacement: it shows that $\mathcal{L}^{grp}_{KP}$ must be abandoned, not that the random oracle H is the only public component that can take over its structural role. This distinction is supported by the primary sources themselves: Goyal et al. [2], in the large universe construction (Section 5), introduce a deterministic public function $T(X)$, defined exclusively from elements published at $Setup$, without any random oracle idealization, which plays the symmetric algebraic role to the one played later by H in [3]; the same authors explicitly note, in a footnote, that replacing $T(X)$ with a hash function modeled as a random oracle is a choice motivated by efficiency, not a structural necessity. Since Sahai and Waters are co-authors of both [2] and [3], the choice of H in the CP-ABE construction falls into a design pattern already documented by the same authors for the symmetric algebraic role in KP-ABE: an efficiency choice, not a forced consequence of the translations (T1) and (T2).

5.3.5 Central Theorem: The Shift in the Origin of the Secret

Theorem 5.10 (Shift in the origin of the secret) *Let Π_{KP} be the [2] scheme and $\mathcal{S}$ the transformation from Definition 5.8. Then $\mathcal{S}(\Pi_{KP})$ coincides, algorithm by algorithm, with the Π_{CP} scheme from [3], modulo exactly three substitutions:*

(i) renaming the Setup secret $y \to \alpha$, preserving its role as the message masking exponent (Proposition 5.7);

(ii) replacing the group component $\mathcal{L}^{grp}_{KP}$ (based on $\{t_i\} \subset MK$) with the group component $\mathcal{L}^{grp}_{CP}$ (based on the random oracle H), a substitution forced by Proposition 5.9;

(iii) introducing the global parameter β and the pair $(C, D) = (h^s, g^{(\alpha+r)/\beta})$ to reconnect α (translated according to (i)) with s (translated according to (T1)), according to (T3).

More precisely, for each algorithm:

$$\mathcal{S}(\Pi_{KP}).Encrypt(M, \mathcal{T}) = \big(\mathcal{T},\ Me(g,g)^{\alpha s},\ h^s,\ \{\mathcal{L}_{CP}^{grp}(x)\}_{x\in Y}\big) \ \equiv\ Encrypt_{CP}(M, \mathcal{T}),$$

$$\mathcal{S}(\Pi_{KP}).KeyGen(S) = \big(g^{(\alpha+r)/\beta},\ \{\mathcal{F}_{CP}(S, r)\}\big) \ \equiv\ KeyGen_{CP}(S),$$

$$\mathcal{S}(\Pi_{KP}).Decrypt = \tilde{C}\big/\big(e(C, D)/Combine(\dots)\big) \ \equiv\ Decrypt_{CP}.$$

Proof 5 *For $Encrypt$: according to (T1), $\mathcal{L}^{poly}(\mathcal{T}, s)$ is executed here (Lemma 5.2 guarantees the identity of the procedure with $Encrypt_{CP}$); according to (ii)/Proposition 5.9, the group component used is $\mathcal{L}_{CP}^{grp}$, producing exactly $\{C_x, C'_x\}_{x\in Y}$ from [3]; according to (T3), $C = h^s$ (the parameter β from (iii)) and $\tilde{C} = Me(g,g)^{\alpha s}$ (with α from (i)) are added. All components coincide syntactically with $Encrypt_{CP}$.*

For $KeyGen$: according to (T2), $\mathcal{F}(S, r)$ is executed here; the necessity of the per-attribute cancellation mechanism (from §5.3.4) produces exactly $\mathcal{F}_{CP}(S, r) = \{(D_j, D'_j)\}$; according to (T3), $D = g^{(\alpha+r)/\beta}$ is added. It coincides syntactically with $KeyGen_{CP}$.

For $Decrypt$: Lemma 5.6 (the Π_{CP} case) provides $Combine = e(g,g)^{rs} = A$; Proposition 5.7(2) provides $\tilde{C}/(e(C, D)/A) = M$, identical to $Decrypt_{CP}$.

The three verifications exhaust the four algorithms ($Setup$ is implicit in (i)+(iii): $Setup_{CP}$ chooses α, β, replacing the choice of $\{t_i\}, y$ from $Setup_{KP}$). ■

5.3.6 Discussion and Implications

Table 5.1 summarizes the correspondence established by Theorem 5.10.

Corollary 5.11 *$\mathcal{S}$ induces an association at the level of* algebraic role *between the components of Π_{KP} and Π_{CP}: each component of Π_{CP} has an identifiable correspondent in Π_{KP}, with the exception of the parameter β and the masking pair (D_j, D'_j), which represent additional components, without a correspondent in Π_{KP}, introduced exclusively to compensate for the loss of direct access to the group component $\mathcal{L}_{KP}^{grp}$ (Proposition 5.9).*

5.3.7 Correctness Transfer vs. Security Transfer under $\mathcal{S}$

Theorem 5.10 establishes an algorithm-by-algorithm coincidence between $\mathcal{S}(\Pi_{KP})$ and Π_{CP}: the same algebraic expressions, the same computational steps, the same Lagrange reconstruction mechanics upon decryption. A question that naturally arises,

Structural role	Π_{KP} **([2])**	$\mathcal{S}(\Pi_{KP}) \equiv \Pi_{CP}$ **([3])**
Location of $\mathcal{L}^{poly}$	$KeyGen$	$Encrypt$
Secret shared by $\mathcal{L}$	y (Setup)	s (fresh, Encrypt)
Location of $\mathcal{F}$	$Encrypt$	$KeyGen$
Exponent encoded by $\mathcal{F}$	s (fresh, Encrypt)	r (fresh, KeyGen)
Group component $\mathcal{L}^{grp}$	$g^{q_x(0)/t_i}$, via MK	$(g^{q_x(0)}, H(i)^{q_x(0)})$, via H
$Combine$ output	$e(g,g)^{sy}$ — direct masking	$e(g,g)^{rs}$ — correction term
Masking exponent	y	α (= y renamed)
New parameter	—	β (links α with s)

Table 5.1: The correspondence induced by $\mathcal{S}$ between Π_{KP} and Π_{CP}.

given this coincidence, is whether the algorithmic identity extends to the level of security guarantees: if Π_{CP} is obtained, at the algorithm level, from Π_{KP} through $\mathcal{S}$, can we state that the security of one is equivalent to, or reduces to, the security of the other?

The answer, supported both by the formal structure of $\mathcal{S}$ and by the security proofs published for the two schemes, is negative. The distinction is precise and deserves to be formalized: $\mathcal{S}$ transports *correctness* (completeness), but it does not transport, by its construction, any *security* guarantee.

Goyal et al. [2] prove the security of the Π_{KP} scheme (the access tree construction) by direct reduction, in the standard model, to the Decisional BDH assumption (Section 2.1.2): an adversary with advantage ϵ in the Selective-Set game induces a simulator with advantage $\epsilon/2$ in the DBDH game, without any idealization of any hash function and without resorting to the generic group model. Bethencourt et al. [3] prove the security of the Π_{CP} scheme through an argument of a different nature: in the Generic Bilinear Group Model, combined with modeling H as a random oracle (Section 2.3.3), showing that the advantage of any generic adversary is $O(q^2/p)$. This proof offers a heuristic guarantee, not a reduction to a standard hardness assumption, a fact that the authors themselves explicitly acknowledge, listing the achievement of a proof under a standard, non-interactive assumption as an open research direction [3].

Moreover, even the authors of the Π_{CP} scheme explicitly reject the possibility that the security technique of Π_{KP} could be directly reused: they observe that, since in their construction private keys are associated with attribute sets (not with access structures), the collusion resistance mechanism based on secret sharing from [2] no longer applies, requiring a completely new private key randomization technique, with a two-level

masking mechanism [3]. In other words, exactly at the moment when Theorem 5.10 shows that the algorithms of the two schemes coincide, the authors of the Π_{CP} source confirm, independently, that the security argument must be rebuilt from scratch.

Proposition 5.12 (Correctness vs. security under $\mathcal{S}$) *Let $\mathcal{S}$ be the transformation from Definition 5.8. Then:*

(1) If Π_{KP} is correct – that is, for any access tree $\mathcal{T}$ and any attribute set γ with $\mathcal{T}(\gamma) = 1$, $Decrypt_{KP}\big(Encrypt_{KP}(M, \gamma),\, KeyGen_{KP}(\mathcal{T})\big) = M$ – then $\mathcal{S}(\Pi_{KP})$ is also correct: for any $S \models \mathcal{T}$, $Decrypt_{CP}\big(Encrypt_{CP}(M, \mathcal{T}),\, KeyGen_{CP}(S)\big) = M$.

(2) $\mathcal{S}$ does not guarantee, by its own definition, any security transfer between Π_{KP} and $\mathcal{S}(\Pi_{KP})$.

Proof 6 *(1) It is the direct consequence of Theorem 5.10: the four components of $\mathcal{S}(\Pi_{KP})$ coincide syntactically with those of $Encrypt_{CP}$, $KeyGen_{CP}$, $Decrypt_{CP}$, and the mechanism by which decryption succeeds, the evaluation of $Combine$ over the satisfied leaves, followed by Lagrange interpolation (Lemma 5.6), is the same algebraic mechanism in both schemes, applied to the same pairs $(\mathcal{T}, X)$ with $X \models \mathcal{T}$. Correctness is, therefore, a property of a single execution of the four algorithms and is invariant under the relocation of the components described by (T1)-(T3).*

(2) $\mathcal{S}$ is defined (Definition 5.8) exclusively as a recomposition of the algorithmic components $\mathcal{L}^{poly}$, $\mathcal{L}^{grp}$, $\mathcal{F}$ and the masking mechanism, evaluated in isolation, for a single invocation of $Setup$, $Encrypt$, $KeyGen$ and $Decrypt$. Security, unlike correctness, is not a property of a single execution, but of repeated adversarial interaction with these four procedures, a polynomial number of $KeyGen$ queries, followed by a $Challenge$ query, according to the games described in Section 2.3, and proving security requires the construction of a simulator that responds consistently to all these queries simultaneously, embedding an instance of the hard problem. $\mathcal{S}$ does not define any mapping between such a simulator for the game of Π_{KP} and one for the game of Π_{CP}; it acts a single time, on the four isolated algorithms, not on the multi-query interactive protocol between the adversary and the challenger. In the absence of such a mapping, no conclusion regarding the security of $\mathcal{S}(\Pi_{KP})$ can be drawn from the simple application of $\mathcal{S}$. This absence is, moreover, confirmed constructively: (T3) introduces the parameter β and the pair (D_j, D'_j), components with no preimage in Π_{KP} (the Corollary above), therefore a new algebraic surface, absent from Π_{KP} and thus not covered by any of its security guarantees.

This distinction extends, through the lens of the transformation $\mathcal{S}$, the observation already made descriptively in Section 3.5 (subsection *Asymmetry in security proofs*): the fact that Π_{CP} could only be proven in the Generic Group Model, whereas subsequent LSSS constructions required the q-Parallel BDHE assumption for security in the standard model, is not an isolated historical coincidence, but a consequence of the fact that the algebraic identity established by $\mathcal{S}$ operates strictly at the level of correctness, without providing any instrument for reducing the security of one to the security of the other. Significantly, when security in the standard model was finally achieved for a CP-ABE scheme, it was not achieved by transporting the DBDH reduction of Goyal et al. through $\mathcal{S}$, but independently, by Waters [13], through a construction based on LSSS and the q-Parallel BDHE assumption (Section 2.1.2), further confirming that the algebraic axis (the subject of this chapter) and the security axis remain, in the actual literature, independent of each other.

5.3.8 From $\mathcal{S}$ to ReKeyGen: The Algebraic Substrate of Delegation

The previous section showed that $\mathcal{S}$ transports correctness, not security. A distinct question is whether the algebraic identity established by Theorem 5.10 has any relevance for the CP-ABPRE architecture from Section 4.2.1, in particular, for the $ReKeyGen(sk_S, S, M') \rightarrow rk_{S \rightarrow M'}$ algorithm, whose first argument is exactly the delegator's private key.

Proposition 5.13 (SK_{CP} as a substrate without preimage in Π_{KP}) *Let $SK_{CP} = \big(D, \{D_j, D'_j\}_{j \in S}\big) = KeyGen_{CP}(S)$, according to Theorem 5.10. Then both components of SK_{CP} belong to the material introduced by $\mathcal{S}$ in addition to Π_{KP}: $D = g^{(\alpha+r)/\beta}$ depends on β, and the pairs $\{D_j, D'_j\}_{j \in S}$ are, entirely, components without preimage in Π_{KP} (the Corollary above).*

Proof 7 *The corollary following Theorem 5.10 lists β and (D_j, D'_j) as the only components of Π_{CP} without a preimage in Π_{KP}. Since D and $\{D_j, D'_j\}_{j \in S}$ exhaust the components of SK_{CP}, and D depends on β through the exponent $(\alpha+r)/\beta$, the entire key SK_{CP} carries, structurally, exactly this material.*

According to Proposition 5.13, the argument sk_S received by $ReKeyGen$, in the Π_{CP} instantiation, consists exclusively of components without a preimage in Π_{KP}. In other words, $\mathcal{S}$ not only reproduces Π_{CP} from Π_{KP} (Theorem 5.10); it also identifies,

without additional effort, exactly the algebraic material that any subsequent delegation mechanism, $ReKeyGen$ included, will have to manipulate.

This identification is strictly of an algebraic nature, not of security: it does not establish any master key security guarantee (Section 4.3.1) for a CP-ABPRE scheme built on this substrate, a property that remains, like the IND-CPA/CCA security discussed in Section 5.3.7, a distinct result, established independently through dedicated proofs [15].

5.4 Obtaining KP-ABPRE and CP-ABPRE

The transformation $\mathcal{S}$ defined in Section 5.2 operates at the level of the basic schemes Π_{KP} and Π_{CP}. A natural question is whether the same structural pattern is found when the two schemes are extended with the proxy re-encryption mechanism, producing KP-ABPRE, respectively CP-ABPRE, as instantiations of the same AB-PRE primitive.

Both constructions follow the same syntactic pattern: the basic ABE scheme is extended with two additional algorithms, ReKeyGen and ReEnc, which allow a proxy to transform a ciphertext without accessing the plaintext. The architectural difference between the two instantiations directly reflects the difference between Π_{KP} and Π_{CP}: in KP-ABPRE, the re-encryption key is generated by a user whose private key contains an access structure, and the re-encryption transforms a ciphertext labeled with attributes; in CP-ABPRE, the re-encryption key is generated by a user whose attributes satisfy the policy in the ciphertext, and the re-encryption produces a ciphertext associated with a new policy.

Proposition 5.13 already identifies that the argument sk_S received by ReKeyGen in Π_{CP} consists exclusively of components without a correspondent in Π_{KP} — namely β and the pairs (D_j, D'_j). In other words, the algebraic material that any delegation mechanism in CP-ABPRE must manipulate is precisely the one introduced by $\mathcal{S}$ in addition to Π_{KP}, not the one inherited from it.

The question of whether there exists a transformation analogous to $\mathcal{S}$ that takes a KP-ABPRE scheme directly to a CP-ABPRE scheme remains open. Such a transformation would have to simultaneously manage the relocation of the sharing procedure and the relocation of the per-attribute randomization component in the presence of the re-encryption key, which introduces additional constraints compared to the case with-

out PRE: the key $rk_{S \to \mathcal{M}'}$ algebraically links components from both schemes, and any relocation must respect this link without exposing the delegator's long-term secrets. This direction is outlined as an open research problem in Chapter 6.

Chapter 6

Conclusion and future work

The present work has traced the relationship between KP-ABE and CP-ABE beyond the intuitive description of "architectural inversion" present in the literature (Section 3.5), asking whether this inversion can be made explicit at the level of the algebraic expressions that define the two schemes. Chapter 5 answers affirmatively, through the transformation $\mathcal{S}$, defined as a composition of three operations (T1–T3) applied to the algorithmic components of the KP-ABE scheme by Goyal et al. [2]. Theorem 5.10 shows that $\mathcal{S}(\Pi_{KP})$ coincides, algorithm by algorithm, with the CP-ABE scheme by Bethencourt et al. [3], modulo exactly three identifiable substitutions, of which one, the replacement of the master key-based group component with the random oracle H, is forced by a locality constraint (Proposition 5.9), not freely chosen.

The most important result to remember from Chapter 5 is, however, the distinction established by Proposition 5.12: $\mathcal{S}$ transports correctness, not security. The fact that $\mathcal{S}(\Pi_{KP})$ syntactically coincides with Π_{CP} offers no reduction between the security guarantees of the two schemes, a fact also confirmed by the published history of the two proofs (reduction to DBDH for Π_{KP}, respectively a proof in the Generic Bilinear Group Model for Π_{CP}). The identity established by $\mathcal{S}$ is, therefore, strictly algebraic, and Section 5.3.8 shows that precisely this algebraic substrate, the components without a correspondent in Π_{KP}, coincides with the material that ReKeyGen must manipulate in the CP-ABPRE architecture from Chapter 4.

Limitations. $\mathcal{S}$ is verified for a single pair of schemes, both with a small universe; the extension to large universe constructions is not supported by the sources, because they use a generator g_2 incompatible with the Setup of Π_{CP} (Section 5.3.4). Also, Corollary 5.11 establishes an association at the level of algebraic role strictly for the pair

(Π_{KP}, Π_{CP}); it does not state that any CP-ABE scheme admits a correspondent through $\mathcal{S}$, nor that $\mathcal{S}$ is invertible in general.

Future directions. An open question, suggested by the diagram in Figure 1.1, concerns the other path in the diagram: can a CP-ABPRE scheme be constructed starting directly from a KP-ABPRE scheme, through a transformation analogous to $\mathcal{S}$ applied not to the basic schemes, but to the constructions with proxy re-encryption? An affirmative answer would close the cycle suggested by the diagram and confirm that the algebraic substrate identified in Chapter 5 structurally propagates also to the level of delegation primitives, consolidating the perspective of AB-PRE as a unitary primitive from which KP-ABPRE and CP-ABPRE derive through symmetric instantiations.

Artificial Intelligence

During the preparation of this work, artificial intelligence tools were used only in an auxiliary capacity. More precisely, they were useful in the initial documentation stage, to identify relevant bibliographic sources and obtain a first orientation in the specialized literature. They were also used occasionally to rephrase certain formulations, for the purpose of clearer expression, without affecting the content or the argumentation. For this purpose, ChatGPT (OpenAI) and Gemini (Google) were used.

All bibliographic sources included in the work were verified and consulted directly, in their original form. The original contributions, the formal construction of the transformation $\mathcal{S}$, the proofs of the results, and the comparative analysis of the cryptographic schemes, were carried out entirely independently, without artificial intelligence tools being involved in generating the mathematical content or elaborating the technical arguments.

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